Molecular cloning, expression and purification of Ballchen kinase domain

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CERTIFICATE OF APPROVAL

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Declaration

The work described in this thesis was carried out in the period from May 2023 to May 2024 at Lahore University of Management Sciences under the supervision of Prof. Dr. Muhammad Tariq at the Department of Biology. I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, contains no material previously published or written by another person neither has it been accepted for the award of any other degree or diploma at a university or any other institute of higher learning, except where due acknowledgment has been made in the text.

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Abstract

In multicellular eukaryotes, differential gene expression patterns established during pattern formation play a major role in cell fate determination. The cell type specific gene expression patterns are transmitted through cell divisions in all cell lineages, often referred to as transcriptional cellular memory. Genetic and molecular analysis in *Drosophila* discovered two evolutionarily conserved groups of genes, Polycomb Group (PcG) and Trithorax group (trxG), which play a pivotal role in maintenance of cell type specific gene expression patterns. The PcG are known to repress and trxG acts as anti-silencers to maintain cell fate. Both the PcG and trxG are known to associate with chromatin and exert silencing or activation, respectively, of their target genes. What are the factors underpinning the phenomenon of gene repression or activation, when both the groups coexist at the chromatin remains elusive? It is thought that such mechanisms are largely governed by cell signaling, which involves post-translational modification of effector proteins. Our laboratory has previously reported that Ballchen (Ball), a known serine/threonine kinase, acts as a trxG regulator. The Ball associates with chromatin at trxG targets across *Drosophila* genome. It is assumed that it antagonizes PcG repression by catalyzing H2AT119 phosphorylation and counteracts PcG mediated ubiquitylation of neighboring residue, i.e., H2AK118ub. However, the detailed mechanistic basis of Ballchen's kinase activity in maintaining epigenetic marks and its role in gene activation remains unknown. This study aimed to understand the molecular basis of gene regulation by Ball. To this end, Ballchen kinase domain was cloned, expressed and purified, which will act as stepping stone to establish a kinase assay. This kinase assay can further be used to identify small molecule inhibitors of Ball, that can then be utilized *in vivo* to further understand the impact of perturbing Ball kinase functioning in genome regulation of Drosophila. The genome of *Drosophila* can then be screened for biomarkers of different cell signaling pathways, in an attempt to understand the signaling pathways involved in Ball kinase function.

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Chapter 1 Introduction and Literature Review

1.1 Drosophila as a model organism

In multicellular eukaryotes, the process of fertilization i.e., fusion of sperm with an egg cell, marks the beginning of development. The zygote that develops as a result of fertilization inheriting genetic information from parents has an inherent ability to give rise to a complete organism. During early embryonic development, after cleavage cell divisions, the process of pattern formation results in formation of different germ layers and body axes. Although different cells contain the same genetic information but as a result of cell-cell communication, cell signaling and the microenvironment of cells within embryo, the process of cell fate determination kicks off. Cells eventually commit to distinct identities based as a result of differential gene expression patterns established during process of pattern formation (Zimmerman & Schultz, 1994; Thompson et al., 1998; Memili & First, 2000 ; Zernika-Goetz, 2003; Santo & Dean, 2004).

We have learned much about the molecular aspects of eukaryotic development using the fruit fly, *Drosophila melanogaster*. In *Drosophila*, immediately after fertilization translation of messenger RNAs deposited by mother during process of oogenesis, also known as maternal effect genes, results in establishment of morphogen gradients across the anterior-posterior (AP) and dorsal-ventral (DV) axes of zygote. The products of maternal effect genes which are transcription factors lead to the expression of the first set of zygotic genes known as gap genes. Unlike maternal effect genes, products of gap genes are expressed in discreet regions along AP axes and gap mutants show loss of major regions in a growing embryo (Nüsslein-Volhard et al. 1987). Like maternal genes, gap genes are also transient and transcription factors and result in activation of pair-rule genes. Expression of pair-rule genes coincide with development of fourteen discreet parasegments which can be visualized by immunostaining with pair-rule genes like even-skiped (*eve*) and fushi tarazu (*ftz*) (Ingham and Martinez 1986). While fly embryo is a syncytium during early (4 hours) embryonic development, pair-rule genes activate segment polarity genes, which define the boundaries of 14 complete body segments and correlates with transition from syncytium to a cellular blastoderm in *Drosophila* embryo (Turner and Mahowald 1977; Foe and Alberts 1983). The cascade of gene expression that kicks off by expression of maternal genes immediately after fertilization finally results in expression of homeotic genes which gives identity to different cells and tissues in growing embryo (Gehring and Hiromi 1986). The conservation of homeotic genes and their expression patterns across vertebrates and invertebrates highlights their role in pattern formation and cell fate determination during evolution. Homeotic genes, such as those found in the Antennapedia (ANT-C) and Bithorax (BX-C) complexes in *Drosophila*, play a crucial role in determining the identity of body segments during development. These genes are often referred to as "selector" genes because they direct the developmental pathway of a particular segment, ensuring that it acquires the appropriate structures and functions. In *Drosophila* and other insects, mutations in homeotic genes can lead to dramatic transformations of body segments. For example, in the Antennapedia (Antp) mutant, antennae are transformed into an additional pair of legs. This demonstrates the role of homeotic genes in specifying the unique identities of segments and ensuring that structures appropriate to each segment are formed.

The clustered arrangement of homeotic genes in the ANT-C and BX-C complexes allows for coordinated regulation of segment identity. By acting as master regulators, these genes determine the fate of each segment along the anterior-posterior axis of the organism. Mutations or alterations in these genes can disrupt this regulatory process, leading to abnormal segment identities and corresponding structural transformations.

Understanding homeotic genes and their role as selector genes provide valuable insight into the molecular mechanisms underlying pattern formation and segment identity during development in organisms like *Drosophila*. (Nüsslein-Volhard and Wieschaus, 1980; Gehring 1985; Gehring and Hiromi 1986; Martinez-Arias, 1988; Gehring et al., 1994; Kaur et al., 2022). The cell type specific gene expression patterns of hometic and non-hometic developmental genes established during early embryonic development are maintained throughout subsequent development. This maintenance of gene expression patterns of cell type specific genes at transcriptional level is referred to as transcriptional cellular memory or epigenetic cellular memory (Bruno 2022). Molecular and genetic analysis in *Drosphila* discovered two groups of genes, namely Polycomb group (PcG) and trithorax group (trxG) genes which maintain transcriptional cellular memory (Kassis 2017). These are evolutionarily conserved groups of proteins which are involved in maintenance of cell fates by regulating gene expression at transcriptional levels. The PcG is involved in maintaining heritable patterns of gene silencing whereas trxG antagonize repressing effect of Pcg and maintains activation of cell type specific genes (Schuettengruber 2017) (Akam et al., 1988; Akam, 1998; Lewis, 1998; Steffen and Ringrose 2014 ;Schuettengruber 2017; Kassis 2017). Both the PcG and trxG proteins act by modifying the local properties of chromatin and several multisubunit complexes of PcG and trxG are known to maintain cell fates and defects in PcG and trxG factors are known to associate with development of disease like cancer (Schuettengruber 2017).

1.2 Eukaryotic DNA packaging

In eukaryotes, the DNA is packaged in the nucleus as chromatin. Nucleosome is the basic unit of chromatin, which consists of approximately 150 bp of DNA, wrapped 1.7 times tightly around an octamer of histone proteins. This octamer comprises two molecules of each of the highly conserved histone proteins, H2A, H2B, H3, and H4. In addition, another histone protein named H1 is, also known as linker histone binds to DNA that exits from histone octamer and links two adjacent nucleosomes. Like beads on a string, nucleosomes are arranged in tandem and lie 30-40 base pairs away (Luger 2001; Ridgway and Almouzni, 2001). The nucleosomal fiber, formed by the packaging of DNA into chromatin, is dynamic and alternates in conformation, throughout the cell cycle. During interphase, chromatin is relatively relaxed, while during mitosis or meiosis, it condenses into distinct chromosomes for segregation.. For instance, metaphase chromosomes comprise densely packed nucleosomal fiber, which is relaxed at least five times as the cell enters the interphase (Manders et al., 2003). While chromatin's structural and functional states alter, such transitions in chromatin states profoundly impact genome regulation (Eberharter a nd Becker, 2002). Such structural perturbations, eventually leading to alteration in gene regulation, are associated with covalent modifications of core histone proteins (Strahl and Allis, 2000).

The core histone proteins, composed of globular domains, are highly conserved. The 'tails' that protrude from the nucleosome are the site of various post-translational modifications (PTMs) (Allis et al., 2007; Berger, 2007; Kouzarides, 2007). Nucleosome composition, chromatin structure, and histone covalent modifications form an important epigenetic layer that profoundly impacts gene expression (Campos and Reinberg 2009). Some of the best-characterized histone tail modifications include methylation, acetylation, and phosphorylation. Meanwhile, some less well-studied ones include ubiquitination, sumoylation, ADP ribosylation, and deamination. Moreover, proline isomerization is a non-covalent modification and is found in histone H3. The amino acids like lysine are the sites of chemical modifications and can accommodate mono, di, or tri-methylation, while arginine can cater to one or two methyl groups. Since histones carry these essential amino acids in abundance, many combinations of such modifications can exist. In specific cellular states and processes, histone variants replace canonical histones, leading to different outcomes. (Henikoff and Smith, 2007).

The dynamic nature of histone PTMs is attributed to the interplay of families of enzymes that catalyze or remove the modifications (Allis et al., 2007; Berger, 2007; Kouzarides, 2007). For instance, acetylation is carried by histone acetyltransferases (HATs) and removed by the action of histone deacetyl transferases (HDACs) (Yang and Seto, 2007). Similarly, phosphorylation observed most commonly on selected serine or threonine residues in histones, is carried out by histone kinases and removed by different histone phosphatases (Rosetto et al., 2012). What factors regulate the action of these histone modifying enzymes and how these enzymes target specific loci are areas of intense research.

Sce (also known as dRing) of PRC1 functions as an H2A ubiquitin-ligase and is essential for depositing the H2AK118ub chromatin mark (Wang et al., 2004). The catalytic activity of E(z) of PRC2 is necessary for Hox gene repression, indicating that the H3K27me3 mark is required for PcG-mediated repression (Cao et al., 2002; Czermin et al., 2002; Müller et al., 2002).

Ash1 and Trx, the members of Trx family of proteins catalyze H3K36 dimethylation, H3K4 methylation, and H3K27 acetylation, are associated with active regions of the genome (Barski et al., 2007; Bell et al., 2007) . Furthermore, these proteins antagonize the PcG mediated repressive marks. For instance, both H3K4 and K36 methylation inhibit the catalytic activity of PRC2 in vitro through an allosteric mechanism (Yuan et al., 2011), and the acetylation of H3K27 directly blocks the methylation of this residue (Tie et al., 2009). In addition, Brahma (BRM), another TrxG gene was shown to activate transcription via ATP-dependent chromatin remodeling (Tamkun et al. 1992).

Another important factor in altering chromatin structure utilizes energy released from ATP hydrolysis. This alters the positioning of nucleosomes. The two families of chromatin remodelers are the SNF2H or ISWI family and the Brahma or SWI/SNF family (Allis et al., 2007). The mode of action of SNF2H/ISWI involves mobilizing nucleosomes along the DNA, while Brahma/SWI/SNF complexes expose DNA bound to histone by altering nucleosome structure. The histone-modifying enzymes and chromatin remodelers interact intimately to alter the gene regulation pattern in each cell at a given time (Schuttengruber 2017).

1.3 Different states of chromatin

In terms of chromatin structure and gene activity, eukaryotic genome is broadly grouped into two states of chromatin. The first is the relaxed genomic environment, which is generally genic-rich and gene-active. This part of the chromatin is called euchromatin. On the other hand, a repressed state of chromatin is seen in cases such as centromeres and telomeres. These highly condensed gene regions are known as heterochromatin which is also known as gene poor region because it lacks protein coding genes. Heterochromatin is further categorized as facultative heterochromatin and constitutive heterochromatin (Trojer and Reinberg, 2007).

Facultative heterochromatin is region of chromosome which is similar to euchromatin but it becomes highly condensed and genes are turned off like heterochromatin at certain developmental stage. So facultative heterochromatin (fHC) refers to the developmentally regulated heterochromatinization of specific regions of the genome. A well-known example is the inactive X chromosome in mammalian females. This inactivation of one of the two X chromosomes in human females involved a large non-coding transcript known as Xist as well as presence of H3K27me3 marks and Polycomb repressor complexes (PRCs) (Trojer and Reinberg, 2007). The EZH2 methyltransferase in PRC2 complex catalyze H3K27me on inactive X. It is known that after the H3K27me3 mark is established, PRC2 is also recruited to the sites of DNA replication where PRC2 containing EZH2 then ensures the maintenance of the H3K27me3 mark on these newly replicated loci. This is how the facultative chromatin is maintained once X inactivation is established early during development (Trojer and Reinberg, 2007).

Contrary to facultative heterochromatin, constitutive heterochromatin marks the permanently silenced genomic regions. The constitutive heterochromatin associates with high levels of H3K9me3 which is catalyzed by Su(var)3-9, methyltransferase and this methyl mark at H3K9 position is recognized and bound by HP1α/β protein. To ensure the faithful transmission of this mark, during replication, HP1 binds to the H3K9me2/3 mark and recruits Su(var)3-9, methyltransferase which then adds the same mark to the newly replicated histones to ensure the fidelity of this mark. The presence of a feedback loop in the maintenance of facultative heterochromatin highlights the dynamic nature of the process, which alternates between constitutive and facultative heterochromatin (Trojer and Reinberg, 2007).

Compared to heterochromatin, euchromatin is a region where chromatin is found in relaxed and an open state. There exist differences in euchromatin regions, and specific histone modifications are concentrated in specific regions. For instance, H3K4me1 is found in active transcriptional enhancers (Barski 2007; Hon 2009). High levels of H3K4me3 mark active genes, and this marks the start of the Transcription Start Site (Schneider 2004; Barski 2007). Meanwhile, H3K36me3 is present throughout the transcribed region (Bannister et al., 2005). Studies from yeast have revealed that at the transcription initiation, the serine five phosphorylated CTD of RNAPII serves as the docking site for scSet1 H3K4 methyltransferase (Struhl et al., 2003). During the transcriptional elongation, scSet2 H3K36 methyltransferase interacts with RNAPII CTD at phosphorylated serine 2 (Xiao et al., 2003).

1.4 Histone modifications and gene expression

Although the literature detailing the effects of histone modifications on transcription is complex, studies continue to expand our knowledge in the mechanistic details of the histone mediated gene regulation. The three central principles involved in regulating gene expression through histone modifications include the direct impact of PTMs in chromatin regulation that affects the higher order conformation, the *trans*-effect, where PTMs lead to disruption of proteins that bind to the chromatin. In addition to these two potential mechanisms, PTMS can act as sites for recruiting specific effector proteins to the chromatin. These last two are examples of *trans* effects (Kouzarides and Berger, 2007).

Studies indicate that, generally, acetylation of histones leads to the activation of transcription. One mechanism that explains this is the recruitment of HATs to chromatin by the action of DNA-bound activators (Haldar et al., 2022). In contrast, repressors of transcription involve recruitment of HDACs to deacetylate histones, which correlate with the repression of transcription (Kouzarides and Berger, 2007). Acetylation of lysine residues leads to the opening of chromatin because acetylation results in neutralization of positively charged side chains, reducing the affinity of the DNA-histone complex. This opens DNA-binding sites, allows the recruitment of transcriptional machinery, and encourages transcription (Haldar et al., 2022). In addition to acetylation, methylation of lysine residues in histones has been exceptionally well characterized. For instance, Histone H3 residues K4, K36, and K79 are linked with gene activation, while the residues K9 and K27 on histone H3 and K20 of Histone H4, when methylated, lead to repression of transcription (Kouzarides and Berger, 2007). However, how these modifications are regulated and what signal(s) impact addition or removal of these histone modifications remains poorly understood. Despite presence of diverse cell signaling proteins in cells, little is known about cell signaling to chromatin and only few phosphorylation sites in histones are known. For example, H3S10, H3S28 and H2AT119 phosphorylation marks are known to positively impact gene activation. The addition of phosphate group adds a negative charge to histone and impacts the structure of nucleosomes (Zahiri et al., 2020). However, little is known about how these phosphorylation marks interplay with modifications of others residues and how cell signaling to chromatin impacts decisions to either activate or silence a specific gene.

Molecular techniques, including Chromatin immunoprecipitation (ChIP) that can detect specific PTMs of histones using specific antibodies, together with microarray analysis, have enabled understanding of these modifications in the whole genome (Mikkelsen et al., 2007; Spivakov and Fisher, 2007). This has allowed us to characterize certain modifications correlating with active or repressed gene states. For example, actively transcribed genes have H3K4me3 modification in their promoters, whereas H3K9me3 is mainly present in repressed gene promoters and correlates with constitutive heterochromatin. These marks are also conserved across species and have allowed us to define the signature modifications for different chromatin states (Ozsolak et al., 2008; Won et al., 2008; Guttman et al., 2009; Heintzman et al., 2007, 2009).

1.5 Histone Phosphorylation

Phosphorylation of histones is dynamic and leads to variable effects in the cellular processes, such as in the cell cycle and transcription. Most commonly, the phosphorylated residues are serines, threonines, and tyrosines of the N-terminal of histone tails. The levels of the phosphorylation modification are regulated by the action of kinases that add phosphate group and phosphatases that remove the phosphate moiety (Oki et al., 2007).

Protein kinases catalyze the transfer of a phosphate group from the ATP molecule to the hydroxyl group of the target amino acid of a protein molecule. This adds a negative charge to the histone protein, which affects chromatin structure. While it needs to be answered how kinases are specifically recruited to their site of action on chromatin, one explanation comes from the case of the MAPK1 enzyme. MAPK1, through its DNA binding domain, can be loaded on its target loci (Hu et al., 2009). Alternatively, these enzymes may require other factors before contacting DNA and catalyzing their action. Though in most cases, histone tails are modified, residues within core regions can also be phosphorylated, as seen in the non-receptor tyrosine kinase JAK2, which phosphorylates histone H3 Tyrosine 41 (Dawson et al., 2009).

Relatively little is known about the functioning of histone phosphatases. However, phosphatase activity must exist within the nucleus, given the precise importance of regulating these modifications for cell functioning. For example, PP1 phosphatase antagonizes the H3S10ph and H3S28ph laid down at mitosis by Aurora B (Goto et al., 2002; Sugiyama et al., 2002).

1.6 Cross-talk between histone modifications

Different post-translational modifications on single histones, single nucleosomes, and nucleosomal domains serve as readouts for various cellular processes. Such chromatin modifications define the local and global gene expression patterns (Strahl and Allis, 2000; Turner, 2000). Therefore, chromatin acts as a signal integration hub, and different extracellular and intracellular signaling cascades can influence its structure (Cheung et al., 2000). Moreover, these modifications are tightly regulated and are context-dependent. H3S10 phosphorylation (H3S10p) exhibits a dual role, acting as both an activator and a repressor of gene expression, depending on the cellular context. It is implicated in the activation of transcription, and this modification is also present on highly condensed chromosomes in mitotic chromosomes attached to spindle fibers (Cheung et al., 2000). Furthermore, one covalent modification can lead to an altogether different and sometimes opposite physiological outcome. H3S10p for instance, is tightly regulated with the cell cycle progression. While this histone modification is minimal during interphase, the same modification becomes pronounced during the mitosis and meiosis. H3S10p begins in late G2, spreads across chromatin during prophase and metaphase as chromosomes condense. However, it decreases rapidly during late anaphase and disappears completely upon exiting mitosis (Hooser et al., 1998). This suggests that different combinations of chemical modification on the same locus result in different cellular processes (Turner, 2000; Jenuwein and Allis, 2001).

While serine and threonine residues undergo phosphorylation by kinases, lysine, and arginine, offer different chemical modification options. For instance, acetylation, mono-ubiquitination, mono-, di-, and tri-methylation have been reported in the case of lysine. Arginine residues also undergo mono or di-methylation in symmetric or asymmetric contexts (Zhang and Reinberg, 2000; Bannister et al., 2002). Although it remains to be understood how these choices are made, enough evidence supports that individual residues exhibit specific preferences in accommodating certain modifications. This phenomenon is intimately linked to the transcriptome of a cell. A well-characterized example these modifications are the H3-K9 and H3-K14 residues that can either be acetylated or (mono-, di-, tri-) methylated (Zhang et al., 2002; Zhang et al., 2002). It is also true that different marks like acetylation and methylation cannot coexist at a given instance. Therefore, competition exists amongst these PTMs; one modification might exclude or increase the chances of another modification. For example, an acetyl group must be removed to add a methyl group on a particular lysine residue in specific time and context. Specific proteins which catalyze addition of acetylation, methylation or other modification on specific amino acids in histone are called writers which act as part of a large biochemical complex. There are biochemical complexes containing histone deacetylases (HDACs) and histone methyltransferases (HMTs) which regulate process of transcription in eukaryotes (Czermin et al., 2001; Vaute et al., 2002; Zhang, 2002).

Enzymes that modify the histone tails exhibit specificity towards the pre-existing state of their substrates. This is seen in many cases. For example, the H3S10p dictates the choice of H3K9me (Rea et al., 2000). It is seen in mammalian cells that H3S10p not only inhibits the addition of methyl mark on H3K9 but also precedes and promotes H3K14 acetylation in response to different cell signals (Jenuwein and Allis, 2001). In mammals, MLL (mixed lineage leukemia) protein mediated methylation of H3K4 is stimulated due to acetylation of H3K9 and H3K14 (Milne et., 2002). It coincides with similar modifications on histones of the HOX gene promoters, as seen by the ChIP assays (Milne et al., 2002). Furthermore, there is a cross-talk between different modifications on histone proteins as well, which is also referred to as the 'trans' effect (Wang et al., 2001; Sun et al., 2002; Briggs et., 2002; Drover et al., 2002). The nature of such modifications might not only be restricted to a single nucleosome but can affect larger arrays of nucleosomes or domains. For example, protein P300 which is a histone acetyl transferase (HAT), preferentially acetylates H3 and H4 when K4 is methylated in H3 (Wang et al., 2001). Contrarily, K9 methylation of Histone H3 results in the inhibition of p300-mediated acetylation of Histones H3 and H4 (Wang et al., 2001).

The modified residues on histones, in effect, act as docking sites for specific effectors. This is reminiscent of the interactions in signaling pathways, an example of SH2 domain containing proteins being recruited to the phospho-tyrosines (Pawson et al., 2000). In chromatin associated proteins, bromodomains are integral to many HATs, chromatin remodelers, and TAF250 (general transcription factor) and this domain is known to bind acetylated lysines (Zeng et al., 2002). Recruitment of bromodomain-containing factors to promoters of genes is essential for transcriptional activation (Agalioti et al., 2002; Hassan et al., 2002). On the contrary, chromatin-organization modifier (chromo) domains containing proteins have a binding affinity towards methylated lysines (Rea et al., 2000; Paro et al., 2004). Heterochromatin protein 1 (HP1) binds to H3K9me mark. Similarly, the Polycomb (Pc) protein, which is another gene-silencing protein, is known to bind methylated K9 and K27 in H3 (Cao et al., 2002; Czermin et al., 2002; Kuzmichev et al., 2002). There are many lysine residues in histones which are methylated and/or acetylated and known to regulate transcription but we still need lack a complete understanding of the mechanistic basis of these PTMs. The cross-talk between histone modifications in *cis* and *trans* adds to the system's complexity. Further studies in elucidating the mechanisms of these PTMs and their effects on the genome, therefore, require further scrutiny.

1.8 Homeotic Genes as paradigm for gene regulation during development

Adept three-dimensional development of an animal body across the anterior-posterior, left, right, and dorsal-ventral axes involves cell-to-cell communication, intercellular and intracellular cell signaling, spatio-temporal transcription regulation, and epigenetic programming of the genome. Homeotic genes or Homeobox (*Hox*) genes play a pivotal role in specifying cellular identities and the bilateral development of the body. These evolutionarily conserved sets of genes encode transcription factors (HOX), which act downstream of early effect genes in development (Basson, 2012; Perrimon et al., 2012; Coutelis et al., 2013; Berenguer et al., 2020).

In *Drosophila*, early genetic crosses and recombination experiments revealed *Hox* genes mapped to the 3R chromosome (Gehring and Hiromin, 1986). The genes were found to be present in two distinct clusters, Antennapedia complex (ANT-C) and Bithorax complex (BX-C), each cluster being approximately 300 kb (Lewis 1985). In vertebrates and invertebrates *Hox* follow spatial-teporal collinearity (Papageorgiou, 2015). Towards the turn of the 20th century, massive work on genetics and developmental biology confirmed the presence of Hox in all bilaterians (Ferrier et al., 2000; Kourakis and Martindale, 2001; Wanninger and Wollesen, 2019; Nong et al., 2020). *Hox* genes encode transcription factors, which comprise a domain that contains a helix-turn-helix motif. This domain, mainly called the homeodomain, binds to DNA in a sequence-specific manner (Williams, 1998; Holland, 2001; Holland et al., 2007; Son et al., 2020). One of the essential contributors to the diversity across the animal kingdom is attributed to Hox because of their paramount role in setting up the early body plan of an organism and ensuring appropriate maintenance of cellular identities in the later stages of development (Papagiannouli and Lohmann, 2015; Hrycaj and Wellik, 2016; Domsch et al., 2020).

Understanding the role of *Hox* genes in *Drosophila* has furthered our knowledge about embryonic development. For instance, the homeotic transformation of third thoracic segment (T3) into second thoracic segment (T2) is observed when the genes *abx*, *bx*, and *pbx* are deleted. The *Drosophila* mutants for these three genes fail to develop halteres and instead an extra pair of wings is seen in these mutant flies. Moreover, the posterior thorax transforms into the anterior thorax which resuls in a fly with four wings, compared to a typical wild-type fly with only one pair of wings and halters (Lewis, 1978; Dickinson et al., 1999; Yarger and Fox, 2016).

In flies *Hox* genes were discovered to play an integral role in early embryonic development, proper segment identity, and various cellular processes to ensure normal body physiology. It is essential to ensure proper regulation of their expression. Abnormal expression of *Hox* genes is a cause of different cancers, including melanoma, breast, bone blood, and colorectal cancer (Shah and Sukumar, 2010). Specifically in esophageal carcinoma, the role of HoxA5 and HoxD9 has been reported. HoxA9-13, HoxB13, and HoxC10 are related to the development of cancer in ovaries and prostrate (Cheng et al., 2005; Miao et al., 2007; Zhai et al., 2007). In *Drosophila*, the studies have allowed us to utilize it to study Hox-associated cancer progression.

Overexpression of *Dfd, Ubx,* and *abd-A* in the fat body has revealed these to be leukemogenic (Ponrathnam et al., 2021). Genetic and molecular analysis in *Drosophila* discovered that cell type specific expression profiles of homeotic genes are maintained by PcG and trxG proteins. The process of transcriptional cell memory of homeotic genes by PcG and trxG proteins during development has served as paradigm to understand how gene regulation plays a role in establishment and maintenance of cellular identities during development (Schuttengruber, 2017).

1.9 The Polycomb and Trithorax group of proteins

While using development of extra sex-comb phenotype in flies, Edward Lewis and Pamela Lewis discovered the first PcG gene named Polycomb (Pc). In the later experiments, Edward Lewis observed that mutations in Polycomb transformed anterior embryonic segments into posterior ones due to the ectopic expression of Hox genes (Lewis, 1947, Lewis 1978). The mutant genetic screens that led to the development of phenotypes similar to those seen in Pc mutants were classified into the PcG of genes. Further genetic screens for suppressors of extra sex comb phenotype in *Drosophila* led to the identification of trxG which, when mutated, led to the conversion of embryonic segments into more anterior ones. In simple words, the trxG genes produced phenotypes antagonizing PcG mutations and also result in the loss of Hox expression, hence known as suppressors of PcG (Ingham,1983; Struhl and Akam, 1985; Kennison and Tamkun, 1988; Schuttengruber 2017; Kassis 2017). Significantly, it was determined that these two groups of evolutionarily conserved proteins are essential to maintain the appropriate and precise expression of Hox genes after their initial transcriptional activators disappear. This observation established the PcG and trxG group of proteins as one of the most elaborate and prominent effectors of maintaining epigenetic cell memory. They regulate and maintain the cell fates by antagonizing each other's effect in maintaining the Hox gene expression in the animal body (Ingham, 1985).

1.10 Repression by Polycomb Group Protein Complexes

There are different PcG protein biochemical complexes which are involved in maintaining repression of gene expression (Grossniklaus and Paro, 2014; Schwartz and Pirrotta, 2013). Polycomb repressive complex 1 (PRC1) includes Polycomb (Pc), Polyhomeotic (Ph), Posterior

Sex combs (Psc), and Sex combs extra (Sce or dRing1) as core components in a biochemical complex. The dRing or Sce subunit mono-ubiquitylate histone H2A protein at lysine 118 (H2AK118ub) (Schwartz and Pirrotta, 2013). which is known as a hallmark of PcG repression. This mark is believed to hinder the process of elongation by RNA polymerase II by restricting its recruitment to the chromatin. In addition, Psc and Sce act together to enhance the E3 ubiquitin ligase activity of PRC1. Importantly, Pc, which contains chromodomain, in PRC1 specifically bind the H3K27me3 histone mark and play a crucial role in maintenance of repression. Furthermore, the resulting monoubiquitylation encourages recruiting PRC2 members (Blackledge et al., 2014). The Ph domain allows for self-association and multimerization due to Sterile Alpha Motif (SAM), which motivates PcG binding to targets (Isono et al., 2013). Generally, PRC1-mediated activities lead to chromatin compaction by discouraging the recruitment of activators on the chromatin, and the Psc subunit plays a specific role (Isono et al., 2013). The subunits comprising the PRC2 complex involve p55 (Nurf55 or Caf1), Enhancer of zest [E(z)], Suppressor of zeste 12 [Su(z)12], and Extra sex combs (Esc). The E(z), containing a SET domain, catalyzes the deposition of the H3K27 methylation mark. The addition of this epigenetic mark is made possible by the histone methyltransferase activity of the SET domain contained within E(z) (Pirotta 1998; Muller 2002; Di Croce and Helin, 2013; Schwartz and Pirrotta, 2013). VEFS-box containing Su(z)12 adds to the stability of the complete PRC2 complex and enhances the activity of E(z) (O'Meara and Simon, 2012). On the other hand, the Esc subunit provides scaffolding and facilitates protein-protein interactions via the WD repeats (O'Meara and Simon, 2012). It is reported that Esc enhances PRC2-mediated repression allosterically (Margueron et al., 2009). It encourages preferential binding to E(z) mediated H3K27 trimethylation, resulting in a local spread of silencing (Margueron et al., 2009; O'Meara and Simon, 2012). Although the p55 subunit interacts with Suz(12), its functional relevance remains to be elucidated because its absence has shown no significant impact on PRC2 activity (Nowak et al., 2011; O'Meara and Simon, 2012; Wen et al., 2012).

Besides PRC1 and PRC2 complexes, Pho-containing Polycomb complexes comprise two sub-complexes. The first complex is referred to as the Pho-repressive complex (PhoRc). It contains Pho and Sfmbt. Pho-INO80 is another Pho-containing complex that contains INO80, which carries nucleosome remodeling activity (Klymenko, 2006). Pho is the only PcG protein that contains a DNA binding domain and binds DNA depending on the specific sequence. Pho also helps recruit PcG complexes to specific DNA elements known as PREs but their role in genome regulation needs to be clarified (Grossniklaus and Paro 2014). Earlier genetic screens have shed light on the native nature of PRE as repressive; the activation into the TRE state is followed by opposing PcG-mediated repression by trxG (Klymenko and Müller, 2004; Grossniklaus and Paro, 2014). While the intricate relationship between the two complexes needs to be elucidated further, the progress made in the last century has allowed us to gain an exciting insight into the interplay between the two.

1.11 Activation by Trithorax Group Protein Complexes

In contrast to PcGs, there are several trxG biochemical complexes which are involved in general transcription, epigenetics of gene regulation and chromatin remodeling. Since transcription activation requires multiple steps, trxG factors are broadly grouped into histone-modifying, chromatin remodeling, and DNA-binding proteins (Schuettengruber et al., 2011; Kingston and Tamkun, 2014; Steffen and Ringrose, 2014). TrxG group proteins counteract the function of PcG genes by maintaining the PcG target genes in an active state, and in general, they interact with transcriptional complexes and play a role in transcription activation (Kingston and Tamkun, 2014).

TrxG group of proteins associated with histone-modifying activities include COMPASS, COMPASS-like, TAC1, and ASH1 complexes. These components commonly contain the SET domain HMTs. COMPASS and COMPASS-like complexes containing the HMT subunits which methylate H3K4 residue and contain distinct subunits that have a non-redundant role in cellular functioning. Set1 subunit of the COMPASS complex is involved in the global activation of the genome. In contrast, the specific gene activation is catalyzed by trithorax (Trx*)* and Trithorax-related (Trr) subunits, both of which are a part of COMPASS-like complexes (Schuettengruber et al., 2011; Shilatifard, 2012). The functioning of TrxG complexes is grounded in their structural composition and the subunits they comprise. For instance, a COMPASS-like complex containing the Trx and Menin (Mnn1) targets the activation of Hox genes. In contrast, the complex that involves Trr and Utx subunits targets hormone-responsive genes (Schuettengruber et al., 2011). It highlights that different trxG complexes may also have unique and have specific gene targets which may also be tissue and cell type specific complexes.

On the other hand, the ASH1 and TAC1 complexes play specific roles in antagonizing the PcG-mediated silencing. Both these complexes are involved in the coupling of histone acetyltransferase (HAT) activity via the CREB binding protein (Schuettengruber et al., 2011; Kockmann et al., 2013; Kingston and Tamkun, 2014). TAC1 carries H3K4 methylation and H3K27 acetylation due to Trx and CBP. However, HMT activity of Ash1 catalyze addition of H3K36 methylation which is associated with transcriptional elongation (Schuettengruber et al., 2011). Furthermore, the TrxG comprises complexes that exhibit ATP-dependent chromatin remodeling activity. Such complexes, which include ISWI, SWI/SNF, and complexes containing chromodomain helicase (CHD), read the histone tail modifications deposited by writers containing HMT and HAT activities and activate transcription (Schuettengruber et al., 2011; Kingston and Tamkun, 2014). These complexes, which regulate chromatin structure, contain genes that play a part in cell cycle and cell signaling and their misregulation is a cause of various forms of cancers (Kadoch and Crabtree, 2015; Hodges et al., 2016).

1.12 Recruitment of PcG and TrxG proteins to chromatin

In Drosophila, the PcG and TrG members localize and act on specific DNA sequences termed PRE/TREs) (Grossniklaus and Paro, 2014; Steffen and Ringrose, 2014). These regulatory regions of DNA, which are bistable, play a role in preserving transcriptional memory across the lifespan of development. Extensive studies on the molecular genetics of *Drosophila* have revealed these self-contained regulatory elements to play a pivotal role in the maintenance of 'active' or 'repressed' states of target genes. This initial programming, by the action of PcG and TrxG factors, sets the stage for epigenetic cell memory, which remains intact throughout the life of an organism (Steffen and Ringrose, 2014).

Functional genomics have revealed that the recruitment of PcG and TrxG to their associated PRE/TREs is not limited to only HOX genes. Furthermore, the events in their recruitment to their respective regulatory regions involve much more than just the histone modifications and are are an area of active research. It is, therefore, imperative to understand the molecular and cell signaling events involved in the mechanism of action of these groups of proteins to build an understanding of their role in transcriptional regulation of the eukaryotic genome (Steffen and Ringrose, 2014). In addition, no simplistic model involving specific binding of PcG or TrxG group effectors to PRE/TREs can genuinely depict their binding mode to chromatin since, irrespective of the state of the gene (i.e., being in the active or repressed state), the PcG and TrxG group effectors are found to localize together at same targets (Papp and Müller, 2006; Beisel et al., 2007; Schwartz et al., 2010; Enderle et al., 2011; Kockmann et al., 2013).

In addition to specific cis-acting DNA elements and chromatin modifications, role of long non-coding RNAs (lncRNA) in gene regulation and maintenance of chromatin structure have also revealed their role in localization of PcG and TrxG group proteins on their target loci (Wang et al., 2011; Mondal and Kanduri, 2013; Gomez et al., 2013; Grote et al., 2013; Herzog et al., 2014; Mulvey et al., 2014; Yang et al., 2014; Hamazaki et al., 2015;). Compelling evidence for such mode of regulation is provided by the *Drosophila* PRE/TRE of the vestigial (*vg*) gene. This locus comprises a regulatory element (GEAR box) that produces lncRNAs bidirectionally, and the products are implicated in both gene activation and repression (Herzog et al., 2014). For instance, at the larval stage, transcription of the forward strand of the *vg* locus leads to silencing. In contrast, the reverse strand transcribed at the embryonic stage is associated with gene activation. A positive correlation existed between the forward-strand mediated repression and distant pairing interactions between the PREs. The reverse strand promoted gene activation through inhibiting interaction between the PRC2 and lncRNA, specifically by restricting the HMT activity of the E(z) subunit (Herzog et al., 2014). Similarly, studies have shown the physical interactions between lncRNA and TrxG group components to promote gene expression. A long intergenic noncoding RNA (lincRNA), HOTTIP, (Wang et al., 2011; Yang et al., 2014) was shown to interact physically with the WDR5, a known TrxG protein. A role of HOTTIP in gene activation correlated with the distance of HOTTIP and the surrounding genes. The siRNA-dependent knockdown of HOTTIP led to the observation that genes nearest to the HOTTIP loci were most affected. This was further confirmed by the reduction of the H3K4 trimethylation mark at the target genes (Wang et al., 2011).

1.13 Role of cell signaling in regulation of PcG and TrxG proteins

While it is known that the TrxG-mediated gene activation antagonizes repression carried by PcG proteins at the same gene loci, this simplistic picture it is not entirely true. Different reports have shown an overlap in genomic loci that simultaneously recruit Trx, PRC2, and PRC1complexes (Papp and Müller, 2006; Beisel et al., 2007; Schwartz et al., 2010; Enderle et al., 2011; Kockmann et al., 2013). Furthermore, DNA elements, referred to as 'bivalent domains,' can preserve both the H3K4me3 activation mark and the repressive H3K27me3 mark (Voigt et al., 2013). For instance, it was shown in Kc and Sf4 Drosophila cell lines, using chromatin immunoprecipitation (ChIP) and DNA tilling array techniques, that TrxG and PcG proteins share similar genomic binding sites at the Hox gene containing ANT-C and BX-C loci (Beisel et al., 2007). Further analysis showed that binding sites were shared between PRC1, Pho, and Trx, independent of the gene expression state. However, this result also showed prominent promoter-associated Pho peaks for the genes in a repressed state. Interestingly, the Pho signal was also found at the active genes' loci across the whole genome. This pattern of distribution of Pho across the genome, at both the active and repressed genic regions, indicates switchable gene expression due to these complexes. Furthermore, the interaction of PRC1 with Pol II helps maintain tissue-specific gene expression, and PRC1 also prefers binding to the paused promoters (Enderle et al., 2011; Gaertner et al., 2012). Expansion of ChIP-chip analysis at the BX-C and ANT-C loci with BG3, D23, Sg4, and S2 Drosophila tissue culture cells has demonstrated extensive colocalization of PcG and TrxG proteins (Schwartz et al., 2010; Enderle et al., 2011). These consistent findings show that both complexes act at the same genomic regions.

1.14 Diversity in PcG and TrxG function

PcG and TrxG group of proteins are a versatile group of factors that modulate the genome organization and gene regulation by various mechanisms. Alternative splicing is one of the mechanisms that adds to the repertoire of the PRC1 and PRC2. This results in forming different protein isoforms, which can perform different functions. Specifically, in the case of the PRC2 complex, multiple isoforms for subunits of Ezh1 and Ezh2 have been observed. In post-mitotic cells, for instance, expression of Ezh1β isoform, lacking the enzymatic SET domain sequesters the Eed subunit, not allowing interaction with nuclear Ezh1α and Suz12, thus it inhibits PRC2 mediated repression (Bodega et al., 2017; Brand and Dilworth, 2017). Similarly, splicing of Ezh2 produces isoforms that vary functionally. For example, the exclusion of exon four and partial exclusion of exon 8 create an altered protein complex with different gene targets (Grzenda et al., 2013). Although less well characterized in the case of TrxG proteins, alternative splicing is also a mechanism for achieving functional diversity in TrxG proteins. Ash2L, for instance, undergoes alternative splicing to produce Ash2L1 isoform (80kDa) and Ash2L2 isoform (60kDa) (Xie et al., 2018).

Extrinsic and intrinsic cell signaling cues play a profound role in the modulation of PcG and TrxG proteins, impacting the outcome, i.e., gene repression or activation. Post-translational modifications of the Ezh2 subunit provide compelling evidence for this mechanism. AKT signaling, which results in the phosphorylation of Serine-21 of Ezh2, has been shown to decrease the binding of PRC2 with histone H3, an outcome of which is the reduction in the levels of the H3K27me3 mark. This epigenetic change has been correlated with the uncontrolled growth of cancer cells (Cha et al., 2005). On the contrary, cyclin-dependent kinase CDK1 phosphorylates Threonine-487 of the Ezh2 subunit, leading to a lower methyltransferase activity and promoting mesenchymal stem cell differentiation (Wei et al., 2011). Threonine-367 is yet another target of phosphorylation. Its phosphorylation by p38, as noted in breast cancer cells, translocates the methyltransferase into the cytoplasm. This keeps it sequestered, discouraging its action on the target genes (Anwar et al., 2018).

Alternatively, post-translational modifications (PTMs) of Transcription Factors (TFs) recruiting PcG and TrxG complexes to chromatin can also play a part in modulating their function. For instance, phosphorylation of ETS-related gene (ERG) at Serine-96 has been observed in prostate cells. This modification hinders ERG and PRC2 interaction, thereby relieving PcG-mediated gene repression (Kedage et al., 2017). On the other hand, such PTMs of TFs can also encourage their interactions with PcG and TrxG. For example, Mef2D is phosphorylated by p38, which leads to the recruitment of Trx-COMPASS complexes onto muscle-specific genes (Rampalli et al., 2007). Therefore, external and internal signaling cues can either directly modulate the PcG and TrxG complexes through PTMs or indirectly alter their functioning by modifying TFs that interact with these complexes, impacting the recruitment of PcG and TrxG proteins.

Our lab performed an *ex-vivo* kinome-wide RNAi screen using *Drosophila* cells to identify the putative kinases that maintain cellular memory. The protein kinases catalyzing Ser/Thr phosphorylation were chosen from all the kinases that appeared in primary screen, and after validation in secondary screen Ballchen (BALL) characterized to facilitate trxG in maintenance of gene activation (Khan et al., 2021). Ballchen (BALL), a histone kinase, was previously described to phosphorylate threonine at position 119 in H2A (H2AT119p) adjacent to H2AK118ub mark. First characterized by Aihara et al. 2004, Ballchen, also known as Nucleosomal histone kinase, NHK-1, is a serine-threonine kinase, which mainly phosphorylates H2AT119 in chromatin state. It belongs to a small but highly conserved family of CK1 group kinases, and the conserved portion within the gene is restricted only to the kinase domain across species. It encodes for two transcripts, each translating into a 599 amino acid long chain, 65.9kDa protein. The kinase domain comprises 282 amino acids, and the protein size is 31.302 kDa (Aihara et al., 2004). Our lab demonstrated that BALL-depleted cells show an increased presence of the H2AK118ub1 mark at the transcriptionally active and silent targets of the trxG. Furthermore, ChIP-seq data also revealed that high levels of both Ball and TRX at genomic regions coincide with high levels of H3K27ac and H3K4me3, where little to no PC was found (Khan et al., 2021). Importantly, BALL association at chromatin positively correlated eith presence CBP mediated H3K27ac as well as CBP itself. Interestingly, VRK1 (mammalian homologue of BALL) is known to phosphorylate CREB at ser 133, and this modification can lead to CREB-CBP interaction and upregulation of CREB target genes (Arthur et al., 2014). This may explain the molecular mechanism underlying the cross talk between the BALL and CREB mediated gene activation and maintenance (Shaukat et al., 2021).

VRK1, it is frequently overexpressed in various tumor types. DNA microarray studies to detect the VRK1 expression have detected it's overexpression in breast, lung, head and neck squamous cell carcinomas, colon cancer, liver cancer, gliomas, multiple myeloma, esophageal carcinomas, and hematopoietic tumors. Notably, in breast cancer, a detailed examination of 5681 cases links VRK1 overexpression with increased proliferation, particularly in tumors with poorer prognoses and in estrogen receptor-positive cases (Fournier et al., 2006; Martin et al., 2008; Kim et al., 2013). VRK1 has various downstream targets that include numerous transcription factors, including p53, histones, and proteins involved in DNA damage response pathways. The roles of VRK1 seem to vary depending on the specific cell type and its physiological or pathological condition. It can act as either an oncogenic driver, tumor suppressor, or a gene predisposing to cancer (Valbuena et al., 2008).

NHK-1 is essential for the meiotic cell division. It is required to form prophase one karyosome, metaphase spindle, and polar body. Due to depletion of BALL, a loss in H2AT119p as well as a loss in H3K14ac and H4K5ac was also observed (Ivanovska et al., 2005). This indicates a possible cross-talk between histone modifications associated with NHK-1 function. This cross-talk can be involved in different biological processes, such as in the case of H3S10p, which leads to H3K14ac and results in the transcription of specific genes. This shows the presence of a histone code in the progression of meiosis that depends on NHK-1 (Ivanovska et al., 2005). Additionally, it is reported that aurora B, Polo, and NHK-1 kinases of the cell cycle work in concert to regulate the process of mitosis (Brittle et al., 2007). This is achieved by spatiotemporally regulating the H2AT119p during the cell cycle. This mark is seen most in the centromeric regions when the cell enters mitosis, and towards the chromosome arms, this modification seems to decrease by the action of Polo. Brittle et al. (2007) have demonstrated that the action of Aurora B, containing complexes INCENP, Survivin, and Borealin, plays a role in concentrating this histone modification at the centromere. Furthermore, NHK1 has a role in the progression of meiosis achieved by phosphorylating BAF (Lancaster et al., 2007). BAF, in an unphosphorylated state, keeps the chromosomes tethered to the nuclear envelope and, upon its phosphorylation by NHK1, allows the reorganization of these chromosomes to form karyosomes. It has been shown that NHK1 is a downstream effector of the meiotic checkpoint, whose function is halted during the DSB formation during meiosis. Because of its role in inner nuclear proteins and DNA, its role was eminent in chromatin reorganization. This study, for the first time, showed at the cellular level that both NHK1 and BAF, as its substrate, are required to form karyosomes by detaining chromosomes from the nuclear envelope (Lancaster et al., 2007).

The role of Ball in the self-renewing ability of male and female germline stem cells has also been reported. It was shown that Ball is not an essential proliferation factor. However, the defects in imaginal discs observed due to *ball* null mutation have been attributed to premature loss of stem or progenitor cells (Herzig et al., 2014). NHK1 is also required to condense chromosomes via the interaction with BAF. Linking DNA to LEM-domain-containing inner nuclear membrane proteins is the major mechanism, including the role of BAF in karyosome formation and chromosome detachment from the nuclear envelope. It is necessary to phosphorylate BAF by NHK1 to allow the condensation of chromosomes in mitosis. In addition to NHK1, the authors have reported the role of the NuRD complex, a chromatin remodeler, to be involved in condensation in a similar context. This is because of its histone deacetylase activity, as deacetylation may play a role in this process, and there is evidence for histone modifications cross-talk (Ohkura et al., 2015). An extremely diverse and versatile role played by BALL in cell division, gene regulation and epigenetic cell memory makes it a promising candidate to further characterize its role in cancer. Our long term goal is to find molecular molecules that interfere with BALL kinase activity to discover details of cell signaling to chromatin as well as efficacy of inhibiting BALL as a therapeutic target in cancers.

With the elucidation of the crystal structure of the first protein kinase, PKA, we came to appreciate that the kinases comprise a bilobate structure. The N-lobe comprises 5 beta-stranded sheets with at least one alpha helix, while the C-lobe is primarily devoid of beta sheets and mainly comprises alpha helices (Knighton et al., 1991). At the interface of the two lobes, an ATP-substrate molecule is sandwiched. The phosphates of the ATP lie tucked under the Gly-rich loop and interact with the C-lobe through the action of a cation, such as Magnesium. The lobes are connected with the help of linked sequences called hinge, which also interacts with adenine of ATP via Hydrogen bonding. The two lobes intimately come into close contact with the help of surfaces formed by the activation loop and the C-helix. The activation loop can be phosphorylated and acts as the site for kinase regulation. At the N terminal side of the A-loop lies a conserved tripeptide motif Asp-Phe-Gly (DFG), which is pivotal for alternating conformation of the A-loop. Importantly, DFG in and DFG out are the most well-known conformations. In DFG-in, the A-loop and C-helix interact, and the A-loop lies further away from the C-helix when in the DFG-out state. The active state of kinases is characterized by DFG-in (Arter et al., 2022). Understanding kinase activity and the structure of kinases is pivotal to designing inhibitors for kinases. For example, some inhibitors compete with the ATP. These are classified into three types. Type 1 inhibitors are the small molecules that bind to the active kinase in the DFG-in, C-in state, whereas type I½ inhibitors bind to the inactive DFG-in, C-out conformation of the kinase. Lastly, type II inhibitors are reported to interact with the kinase in its inactive DFG-out conformation (C-in or C-out) (Arter et al., 2022).

1.16 Kinase assay as a tool to identify inhibitors

Protein kinases catalyze the post translational modification of proteins most commonly at serine, threonine and tyrosine residues. This modification has profound implications for various cellular processes. Protein kinases therefore act as an attractive target for drug development and discovery. There are over 30 drugs approved that readily used for cancer treatment by perturbing the protein kinase activity (Fabbro et al., 2015). Kinase assays provide a way to characterize the functioning of a kinase domain. Kinase activity typically involves measuring phosphoryl transfer by detecting phosphorylated products or ATP/ADP ratio changes. One traditional biochemical approach used for this purpose is the radioisotope filtration binding assay. The radioisotope filtration binding assay conducts the kinase reaction using γ-ATP labeled with a radioisotope. The radioisotope-labeled phosphate from γ-ATP is transferred to the kinase substrate during the reaction. After the reaction, the unincorporated radioisotope is removed through binding and washing steps, leaving only the phosphorylated substrate bound to a filter. Detection of the radioisotope-labeled phosphate incorporated into the kinase substrate is then achieved using autoradiography or scintillation counting. The intensity of the signal corresponds to the level of kinase phosphoryl transfer activity, providing a direct measurement of kinase activity (Haubrich et al., 2016).

Enzymatic assays for kinases serve as a valuable tool for drug discovery by allowing screening for small molecule inhibitors of specific kinases. While kinases may be activated by upstream activators, one of the common methods of their activation is auto-phosphorylation, which typically occurs on the activation loop of a kinase. This change, which can either occur in *cis* or *trans* manner, alters the conformation of a kinase, thereby allowing to interact with its downstream substrates (Hanks et al., 1988; Petsko et al., 2003).

While setting up a kinase assay it is imperative to incubate the kinase with a known or putative substrate, that allows gauging the functionality of the assay. To this end generic substrates have been used, that can be used with multiple kinds of kinases. These include myelin basic protein (MBP), histone isoforms, and synthetic peptides such as kemptide (sequence: LRRASLG), Syntide-2 (sequence: PLARTLSVAGLPGKK), Crosstide (sequence: GRPRTSSFAEG), and CREBtide (sequence: KRREILSRRPSYR) (Cross et al., 1995; Perck et al., 2006). For kinases that are newly discovered and lack any known orthology, as well as show absence of auto-phosphorylation or activity with generic substrates, libraries of proteins or peptide arrays can be screened (Schutkowski et al., 2005).

While there are less reports in identification of small molecule inhibitors for trithorax group of protein, inhibitors of Polycomb group of proteins have been identified. For instance, overexpression of EZH2, SUZ12, and EED is correlated with the progression of various types of tumors. EZH2 is tied to alteration of transcriptional landscape during carcinogenesis as observed across malignant neoplasms. It leads to an increased H3K27me3 levels (Bracken et al., 2003; Okosun et al., 2014). Myelodysplastic syndrome or myeloproliferative neoplasms have been linked to loss of function mutations in PRC2 subunits, including EZH2, SUZ12, and EED (Bejar et al., 2011). Overexpression of PRC2 leads to an increased proliferation of cells. In transgenic mouse models, somatic deletions of EZH2 and EED interact with the Q61K mutation of the NRAS oncogene, resulting in hyperactivation of the STAT3 signaling pathway and leading to acute myeloid leukemias (Dannis et al., 2016).

Aims and Objectives

Our long-term goal to discover molecular inhibitors for BALL will add to list of molecules which may be characterized for modulation of PcG/trxG activities linked to cancer. To this end, in this dissertation I aimed to purify BALL kinase domain which may serve as stepping stone to establish BALL in vitro kinase assay. Although the role of BALL in regulation of gene activation by trxG was established by our lab previously, the detailed mechanism remains unknown.

In my Master's thesis, I aimed to accomplish following specific objectives:

- Develop cloning strategy for N and C term BALL kinase domain
- Molecular cloning of N and C term tagged BALL kinase domain
- Express and purify recombinant BALL kinase domain in bacteria

Chapter 2 Materials and Methods

2.1 Confirmation of pENTR-D-TOPO-Ball

The pENTR™D-TOPO-Ball was acquired from the epigenetics laboratory which was constructed by a PhD student Haider Khan (Khan 2020; Khan et al., 2021) and confirmed by restriction digestion using *BglII* (Thermo Scientific) (Appendix 1). The PCR tube containing reaction components was incubated at 37°C for 60 minutes in thermocycler. After the digestion, the complete volume (20 µL) of the digested product was run against 1kB DNA ladder as well as compared with uncut plasmid. The 1% agarose gel was run at 95 Volts for 50 minutes.

2.2 Competent cells preparation

The frozen *E.coli* stock of stellar cells (gift from Dr. Shoaib) from –80°C were allowed to thaw for a few minutes on ice. Using a pipette tip, the cells were streaked on LB agar plate and this plate was incubated at 37°C overnight. Next day, distinct colonies visible on plates were inoculated in LB broth as a primary culture and incubated at 37°C in a shaking incubator (180 rpm) overnight.

After a minimum of twelve days growth 1ml of the primary culture was added to 50 ml of fresh LB broth and incubated at 37°C in shaking incubator (250 rpm) for 2 hours. Absorbance of growing culture was measured at 600nm using spectrophotometer (eppendorf BioPhotometer). At an O.D of 0.4, the cells were moved to 50 ml falcon tube and were harvested by centrifugation at 4000 rpm at 4°C for 20 minutes. After removing supernatant, freshly prepared $CaCl_2$ (0.05M) was added. These cells were spun (4000 rpm) in a centrifuge at 4°C for 20 minutes and supernatant was removed. The cells were resuspended in 2.5ml of 15% glycerol prepared in 0.1M $CaCl_2$. 100 µL aliquots were prepared from this solution and snap frozen in liquid Nitrogen. The aliquots were stored at –80°C.

2.3 Transformation of pENTR-D-TOPO-Ball into competent cells

For transformation of vector pENTR™D-TOPO-Ball into stellar competent cells, 200 ng of the amplified plasmid DNA was transformed. To eppendorf containing 100 µL stellar competent cells, 2µL (100 ng/µL) plasmid DNA was added and the eppendorf containing stellar cells was tapped gently. Then this eppendorf was left on ice for 30 minutes, which was followed by a 45 seconds heat shock at 42°C in thermomixer. After the heat shock, 500 µL LB broth was added and the eppendorf containing transformed cells was kept at 37°C shaking incubator for 2 hours at 180 rpm. The eppendorf was centrifuged at 6000 rpm for 5 minutes. After discarding the most of supernatant, using a micropipette the remaining cells were collected and spread on LB agar plate containing kanamycin antibiotic (100 µg/ µL). The cells were spread on the plate and incubated at 37°C incubator overnight.

2.4 Mini prep for pENTR-D-TOPO-Ball

From the agar plate containing pENTRD-TOPO-Balll plasmid transformed cells, one colony was picked with a micropipette tip and inoculated in 5 mL LB broth in a test tube containing 5 µL kanamycin antibiotic. This test tube was incubated overnight in a shaking incubator (180 rpm) at 37°C at. Cells were harvested by centrifuging the cultureat 4000 rpm for 20 minutes. The pelleted cells were suspended in 250 µL of the Resuspension solution (Appendix 3). To this tube, 250 µL of the Lysis Solution was added and the solution was mixed thoroughly by inverting the microcentrifuge tube gently, until the solution was clear. Next 350 µL of the Neutralization solution was added and the mixture in microcentrifuge tube was immediately mixed thoroughly by inverting the tube. This was followed by centrifugation at 14000 x g for 10 minutes using the table-top centrifuge at room temperature. The supernatant was carefully moved to the GeneJET spin column (Thermo Fisher Scientific), without the transfer of white precipitate. Then centrifugation was carried out for 1 minute at 14000 x g, with column in the same collection tube. The flow through was discarded and 500 µL of the Wash solution was added to the spin column and the column was centrifuged for 60s at 14000 x g and the flow through was discarded again. This step was repeated once again, with 800 µL Wash solution, the flow-through was discarded and the column was centrifuged for 1 minute at 14000 x g. In the last step, the column was moved to a fresh 1.5ml microcentrifuge tube and 30 µL of elution buffer was added carefully to the center of the column. After incubating for 2 minutes, the column was centrifuged for 2 minutes at 14000 x g. The purified plasmid DNA was collected as the flow-through and stored at –20°C. The concentration of plasmid DNA was measured using DNA nanodrop (Thermo Fisher Scientific) and 2µL of the sample was run of 1% agarose gel for 50 minutes at 95 Volts. The protocol was carried out following the instructions mentioned in GeneJET Plasmid Miniprep Kit Cat#K0502, #K0503 (Thermo Fisher Scientific).

2.5 Primer design for Ballchen kinase domain

The primers were designed for Ballchen kinase domain that incorporated *BamHI* and *HindIII* restriction enzyme sites (Appendix 4) in addition to leader sequences. Furthermore, the primers contained nitrogenous bases that encoded for histidine amino acid (6X CAT codons). The classic restriction digestion based cloning strategy was adapted to add His tag at N terminus and C terminus of the ballchen kinase domain. To achieve this separate pairs of forward and reverse primers were designed, that ensured His tag at N terminus and C terminus in particular sets (Appendix 4). The primers were designed and verified *in silico* using SnapGene.*).*

2.6 PCR for ballchen kinase domain coding sequence

PCR reaction with *Taq Polymerase* was carried out (Appendix 5), initially, with predicted annealing temperatures (58°C, 59°C, 60°C, 65°C) and altering the elongation time (30s and 1 min) for the different primer sets (that was calculated using online NEB Tm calculator tool). 10 µL of each sample was mixed with 6X loading dye (Appendix 6) and loaded in an agarose gel. The gel was run with 3 µL of 1kB DNA ladder (GeneRuler 1 kb DNA Ladder). After each PCR reaction, the results were visualized on the 1% agarose gel run for 50 minutes at 110V.
After optimizing ideal Tm for annealing, the final reaction was carried out with *Phusion Polymerase* (Thermo Fisher Scientific) at 58C with elongation time of 1 minute (Appendix 7) . 5µL of sample mixed with 2µL of 6X loading dye was resolved on agarose gel which was run against 3µL of 1kB DNA ladder (GeneRuler 1 kb DNA Ladder). The gel was run at 110V for 50 minutes.

2.7 PCR purification for pooled PCR products

PCR was carried out in duplicates for each pair of primers amplifying Ball kinase domain coding sequence to generate N and C term His tagged Ball kinase domain following the same protocol as used for amplification using *Phusion Polymerase* (Appendix 7). After the PCR, 2µL of each sample mixed with 2 µL 6X loading dye, was run on 1% agarose gel which was run against 3µL of 1kB DNA ladder (GeneRuler 1 kb DNA Ladder). The gel was run at 110V for 50 minutes.
In the next step, the PCR mixture, of each primer pair, was pooled and collected into respective labelled eppendorfs. The pooled mixture contained within these eppendorfs was purified following manufacturer’s instructions using PCR Purification kit (QIAGEN- QIAquick PCR & Gel Cleanup Kit (cat. nos. 28104,

28106, 28506 and 28115). After the PCR purification, 2μL from each of the eppendorfs was used to check quality of DNA using the nanodrop and 2 μL of each sample with 2μL of 6X loading dye was loaded on agarose gel which was run against 3μL of 1kB DNA ladder (GeneRuler 1 kb DNA Ladder). The gel was run at 110V for 50 minutes.

2.8 Confirmation of pET21-a(+) Vector

The pET21a (+) expression vector was confirmed using restriction digestion enzyme BamHI-HF (New England Biolabs) (Appendix 8). The PCR tube containing reaction components were placed at 37°C for 30 minutes in a thermocycler. After digestion, 25 μL of the restricted sample mixed with 5 μL of DNA loading dye was run on agarose gel. In a separate well, 5 μL of uncut plasmid was also run on the same gel. The samples were run against 1kB DNA ladder. The gel was run for 50 minutes at 80 Volts.

2.9 Midiprep for pET21a(+)

After confirmation of the pET21a (+) vector, it was transformed into competent cells as described in section 2.3. The colonies were picked up and were used to set up 100 ml LB broth culture that contained 100 μg/ μL Ampicillin antibiotic. This primary culture was set up, following the same conditions as earlier (section 2.2), and the cells were harvested the next morning to proceed with the midiprep protocol. The cells were harvested by centrifugation at 4000 rpm for 30 minutes from the overnight grown culture at 37°C shaking incubator at 180 rpm. Purified plasmid DNA was isolated using kit following manusfacturer's instructions (The protocol was carried using PureLink HiPure Plasmid Filter DNA Purification Kits Catalog numbers K2100-14, K2100-15, K2100-16, K2100-17, K2100-26, K2100-27).

2.10 Molecular cloning of Ball Kinase domain in pET21a (+) vector

Double digestion with restriction enzymes *BamHI* and *HindIII* was carried out for each of the N and C His tagged Ball kinase domain inserts. Furthermore, the expression vector pET21a (+) was also double digested with the same set of enzymes. This ensured the generation of sticky ends compatible for the insert and vector ligation. For digestion, 1 µg DNA was used of each of the inserts (PCR amplified coding sequences to generate N and C tagged kinase domain) and pET21a (+). For details, see Appendix 9.. To each PCR tube was added the restriction digestion mixture including the buffer and restriction enzymes, the respective PCR insert. In a separate PCR tube pET21a (+) vector was also treated for restriction digestion (Appendix 9). The PCR tubes were placed in thermocycler for 1 hour at 37°C.

2.11 Gel extraction of the digested Ball Kinase domain insert and pET21a (+)

After the restriction digestion 0.8% agarose gel was run at 100V for 50 minutes. While visualizing the agarose gel on UV trans illuminator restricted DNA-N-His Ball Kinase (846 bp), C-His Ball kinase (846 bp) and pET21a (5443 bp) were excised from the gel using scalpel. The slices of gel containing respective DNA were placed into labelled eppendorfs which were weighed. Extraction of DNA from the gel slices was carried out using Gel Extraction kit following manufacturer instrucxtions (PureLink Quick Gel Extraction Kit Catalog numbers K2100-12 and K2100-25). Finally, the purified DNA in the elution tube was stored at 4°C for immediate use or at −20°C for long-term storage. The concentration of plasmid DNA was measured using the DNA nanodrop.

2.12 Ligation of the restricted DNA

The DNA extracted from gel was utilized to perform ligation using T4 DNA ligase (Thermo Fisher Scientific). Each of the two inserts, N and C His tagged ball kinase, was ligated with the expression vector pET21a (+). The ligation reaction was set up in PCR tube at 16°C in thermocycler, overnight (Appendix 10). The standard 3 :1 molar ration for insert to vector was used. The following formula was used to calculate the amount of DNA required for ligation:

Insert mass in ng = [(ng of vector) **x** (kb size of insert) **/** (kb size of vector)] **x** (molar amount of insert) **/** (molar amount of vector)

Size of Ball Kinase = 846 bp

Size of vector pET21 a (+) = 5443 bp

After overnight incubation of ligation reaction at 16°C , 10 µL of the ligation mix for each of the constructs (N and C His-tagged Ball kinase) was transformed into competent cells. The transformation was carried out as mentioned in section 2.3. The transformed cells were plated on LB agar plates containing Ampicillin antibiotic (100ug/µL). The plates were labelled and placed at 37°C incubator overnight.

2.13 Colony PCR for Ball kinase

The plates were observed for growth of colonies and individual colonies were picked to perform colony PCR. To the PCR tube, each colony was inoculated using micropipette tip. For each tube, 10 μL PCR water was used, and the colony picked was thoroughly mixed in the PCR water. The PCR tubes were then placed at 95C in thermocycler for 5min. Then the tubes were centrifuged at 10,000 rpm for 10 minutes. 1 μL supernatant was used as the template, and PCR was carried with the respective primer sets using *Taq Polymerase*. See appendix 11 for the reaction details. The PCR reaction was resolved on 1% agarose gel for 50 minutes at 110V to confirm the presence of cloned Ball kinase coding sequence.

2.14 Midi prep for PCR confirmed colonies

In the next step, plasmid was amplified through midiprep for the colony PCR confirmed N and C His-tagged Ball kinase domain constructs using PureLink HiPure Plasmid Filter DNA Purification Kits. To achieve this, the PCR confirmed colonies were picked from the LB agar plate and inoculated in 100 ml LB broth culture containing Amp (100 μg/μL). This flask was left at 37°C shaking incubator, 180 rpm for overnight growth of the bacteria. The next day, the culture was collected in 50 ml falcon tube (in two successions) and cells were harvested by spinning at 14000 rpm for 15 minutes in a centrifuge. The pellet was then used to proceed with the Midi-prep, following the manufacturer's instructions as mentioned in section 2.9.

Following this, DNA nanodrop was used to measure the concentration of purified. The plasmid DNA from each colony was then further treated with restriction enzymes, to confirm the presence of insert and validate recombinant plasmid. 1 μg of each of the inserts was used to carry restriction digestion (Appendix 12). The plasmid DNA was sent for sequencing to ensure it did not contain any mutations.

2.15 Expression of N and C His tagged Ball kinase domain in BL21-DE3(RIL)

The confirmed plasmids of N and C His-tagged Ball kinase were transformed into BL21-DE3(RIL) cells. For each construct 200 ng of plasmid DNA was transformed following the same protocol as mentioned in section 2.2. The LB Agar plates containing Amp (100 μg/μL) were streaked with colonies carrying either N or C tagged constructs and incubated at 37°C overnight. Next day, the plates were seen for the growth of colonies. For each plate, colony was picked and inoculated into 5 ml LB broth containing 5 μL Amp (100 μg/μL). This was the primary culture. Next day, in 20 ml LB broth, containing 20 μL Amp (100ug/μL), 2μL of the primary culture of each of the construct was inoculated. This flask was placed at 37C in a shaking incubator, and the cells were grown until an OD_{600} was reached between 0.4 to 0.6. After the OD was between 0.4-0.6, the culture was induced with 1mM IPTG. In addition to the induced samples, one flask was set up as a control for uninduced sample. The cultures were taken out from the shaking incubator after 4 hours of IPTG induction and cells were harvested. The cultures from the flask were moved to 50 ml falcon tubes and centrifuged at at 4000 rpm for 20 minutes. The supernatant was discarded, and the pellet was used to isolate proteins.

The cells collected in falcon tubes were taken for sonicated using ultrasonic cell crusher (BIOBASE). To each of the falcon tubes, 3 ml of 50mM Tris-HCl pH 8.0 was added. Samples were subjected to sonication with 20s pluse ON, while 10s for pulse OFF, and each sample was treated for a total of 20 minutes. Importantly, sonication parameters were set as 30% power rate, an alarm temperature set at 20°C to prevent overheating, a probe temperature of -5°C for sample cooling, and a power supply of 500W at 220V/50Hz. The samples were kept on ice throughout the procedure. These settings facilitated controlled sonication, minimizing sample damage while achieving efficient disruption for various applications, such as cell lysis. After the sonication, the falcon tubes were centrifuged at 4000 rpm for 40 minutes. After centrifugation, supernatant was collected in labelled falcon tubes. The remaining pellet fraction in the falcon tube was proceeded further by addition of 3 ml of 50 mM TrisHCl pH 8.0. The pellets were thoroughly suspended using micropipettes. The supernatant and pellets were further used to prepare SDS PAGE samples, as mentioned in section 2.18.

2.17 Bradford Assay for BALL Kinase Protein

The concentration of the BALL kinase domain protein was measured using the Bradford assay. Initially, a 1:5 dilution of the 5X Bradford Reagent (Bio-Rad, Cat# 500-0006) was prepared with MilliQ water. Subsequently, 200µl of this diluted Bradford reagent was dispensed into each well of a 96-well plate. Following this, 2µl of the protein sample was added to each well containing the diluted Bradford reagent, with each sample run in duplicate to ensure reliability. The 96-well plate was incubated for 5 minutes to allow for the binding of the Bradford dye to the proteins present in the samples. After the incubation period, the absorbance of the protein samples was measured at 595nm using a spectrophotometer. The Bradford reagent with no sample, served as a blank to account for background absorbance. Finally, the concentration of the protein in each sample was determined by comparing the absorbance readings to a standard curve prepared using known concentrations of a protein standard, thereby enabling the calculation of the protein concentration in each sample.

2.18 Sample preparation for SDS-Gel

From each of the labelled falcon tubes containing sonicated cell product, 75µL of sample was taken with the help of micropipette and moved to labelled Eppendorf. 25 µL of 4X lamelli buffer containing DTT (200 mM) was added to each eppendorf containing respective sample. These eppendorfs were then moved to thermomixer which was set up at 95°C and incubated for 10 minutes. The prepared samples were then run on SDS gel (Appendix 13). The gel was run for 30 min at 80V and then 100V for another 4 hours. T SDS-PAGE gel was placed in a plastic tank and coomassie staining solution was added onto the gel and was left for 30 min shaking at room temperature. After removal of staining solution, destaining solution was added. After 30 min the destaining solution was discarded and fresh destaining solution was added to the tank to visualized the gel next day.

2.19 Large Scale Protein production

After verifying the expression of Ball kinase domain in mini-culture i.e., 20 mL culture volume, large scale protein production was carried out using 100 mL LB media. The colony for transformed construct was picked up and inoculated in 100 mL culture, which was incubated at 37 °C at 180 rpm shaking incubator overnight. Next day, the culture was moved to 50 mL falcon tube and cells were harvested by centrifuging at 4000 rpm for 20 minutes, the supernatant was discarded, and the remaining 50 ml culture was added to same falcon tube. This was followed by centrifugation for another 20 minutes at 4000 rpm. The supernatant was discarded and to the pelleted cells was added 5 mL of 50 mM Tris-HCL. Finally, cells were sonicated as mentioned in section 2.15, but the process was carried out for 30 minutes for each of the sample. After centrifugation, supernatant was collected in labelled falcon tubes of each of the constructs (N or C His Ball Kinase). The remaining pellet in the falcon tube was proceeded further by addition of 3 ml of 50 mM Tris pH 8.0. The pellets were thoroughly suspended using micropipettes and used to prepare SDS-PAGE samples following the protocol mentioned in section 2.18.

2.20 Ball Kinase Purification

Before proceeding with the pellet obtained from cells grown in 100 ml LB culture, fresh buffer B (Appendix 15) was prepared that contained benzonase (250 unit/µL) and lysozyme (1% w/v). A total of four 15 mL falcons were used and to each of the falcon tubes 3 ml of resuspended cells were added in prepared buffer B. Three of these falcon tubes were for induced samples and one for un-induced control. All induced samples were collected in one 50 ml falcon tube and the un-induced sample remained in 15 ml falcon tube. Both the falcon tubes were then placed on a shaker for 45 minutes at room temperature to lyse the cells. 100 µL from each sample was collected in Eppendorf tube labelled as WCL (whole cell lysate) for induced and un-induced samples. The remaining samples were centrifuged at 4000 rpm at 4°C for 45 minutes. The falcon tubes containing the remaining samples were placed in -20°C freezer.

2.21 Purification with Ni-NTA column

To 15 ml Ni-NTA coloumn (QIAGEN), 1.5 ml affinity gel was added and allowed to dispense off the ethanol for 3 minutes. 10 ml fresh MiliQ water was poured into the column to wash it. In the next step, 10 ml of Buffer B was added to column and allowed to drain by gravity. Next, all the induced supernatant sample, contained in falcon tube was added to the column. The column was shifted to a new falcon tube, labelled as W1 and 10 ml buffer B was added to column. The flow through in falcon was stored at -20°C. This was repeated subsequently with 10 ml Buffer C, twice with 3 ml Buffer E, into labelled falcon tubes as Wash 2, Elution 1 and Elution 2, respectively. The composition of buffer solutions mentioned above are described in Appendix 15. The elution sample was collected with falcon tube kept on ice. Finally, the column was washed with 6 mL MiliQ water followed by 10 ml 6M Guanidine HCL and finally with 6 ml 30 % ethanol. Before applying ethanol, the column stopper was put on, parafilm coveredcolumn was stored at 4°C. The collected samples were then used to prepare samples for SDS gel. The samples were made as described earlier in section 2.18.

2.22 Dialysis for BALL kinase domain

The fractions collected from elution 1 and 2 were collected and transferred to a 10 cm by 10 cm cut dialysis tubing packet, that was clipped after pouring eluate in respective packets. The dialysis tubing used was of Sigma Aldrich catalogue D9652. After packing elution 1 and 2 in respective packs, the packets were moved into a 2 Liter beaker, which had approximately 200 mL buffer E. Then the beaker was moved to 4°C, and then the folding buffer was poured into the beaker, drop wise, with the help of tubing that was clipped to ensure drop wise flow of the folding buffer (Appendix 16).

Chapter 3: Results and Discussion

3.1 Strategy for molecular cloning of Ball Kinase domain in pET-21a (+)

To begin with the establishment of an *in-vitro* kinase assay for *Drosophila* Ballchen (BALL) protein, it was necessary to have in hand the specific region coding for kinase domain, which could then be set up for confirmation of a functional kinase domain. To this end, nucleotide sequences encoding BALL kinase domain were cloned in pET-21a (+) vector to produce N and C term Histidine (His) tagged BALL kinase domain in bacteria.

The kinase domain off BALL is between 47-328 amino acid residues (UniProt Q7KRY6·NHK1_DROME). This accession was subsequently used, and the corresponding nucleotide bases were found in BALL cDNA sequence available on fly base (flybase.org). An online *in silico* molecular cloning tool, Benchling (https://www.benchling.com/) was used to confirm this sequence encoding amino acids in Kinase domain. The amino acids encoding Kinase domain (846 base pair sequence) was amplified using BALL cDNA clone (Figure 3.1) as a template in a PCR (Khan et al., 2021). While designing PCR primers (see section 2.5 for primer design) for amplification of Kinase domain encoding nucleotides, it was ensured that cloning of this PCR product in pET-21a (+) vector will result in an in-frame cloning with the 6X His-tag sequence to generate a fusion protein. Since multiple cloning sites present in pET-21a(+) facilitate production of C term fused proteins, the molecular cloning strategy to produce N term tagged ball kinase domain PCR primers were designed in such a way that nucleotide sequences encoding 6XHis tag were added into the forward primer. Similarly a a stop codon was incorporated into the reverse primer to ensure it that native His tag in vector at the C terminal does not get translated. For the C terminal tagging of BALL kinase domain with His tag, the forward and the reverse primers lacked any nucleotide sequences encoding 6X His tag sequence. Rather, the reverse primer was designed to ensure that there was no stop codon and the 6X His tag sequence from the vector was in-frame to generate a fusion protein (see Appendix 16 for primer sequences). While designing the primers and incorporating restriction enzyme sites, it was carefully checked that the insert does not contain the same restriction site. This was important because otherwise the restriction enzymes could cleave from within the insert sequence. To this end the nucleotide sequence was uploaded onto the NEB cutter online tool, and sequence analyzed for restriction enzyme sites (https://nc3.neb.com/NEBcutter/). To amplify ball kinase domain sequence, a GATEWAY ENTRY clone (*pENTR-D-TOPO-ball*), constructed previously by a PhD student in our lab (Khan et al., 2021), containing full length BALL sequence was used as a template However, first BALL ENTRY clone was confirmed through restriction digestion with BglII restriction enzyme. Since there are two restriction sites for BglII, an expected fragment of 844 bp was released along with 3536 bp vector sequences which confirmed ball full length sequence containing plasmid (Figure 3.1) fragment. After confirmation, this plasmid was used as a template DNA for amplification of the Ball Kinase domain using the specific primer sets as described above, incorporating His Tag at the N and C terminal end (Details in sections 2.3, 2.5 and 2.6). Similarly, pET-21a (+) expression vector was also confirmed via restriction digestion with Bam- HI (HF) enzyme. Since there is a single restriction site for Bam HI in pET-21a (+), digestion with Bam HI revealed a linear plasmid of expected 5443 bp (Figure 3.2).

(A) (B)

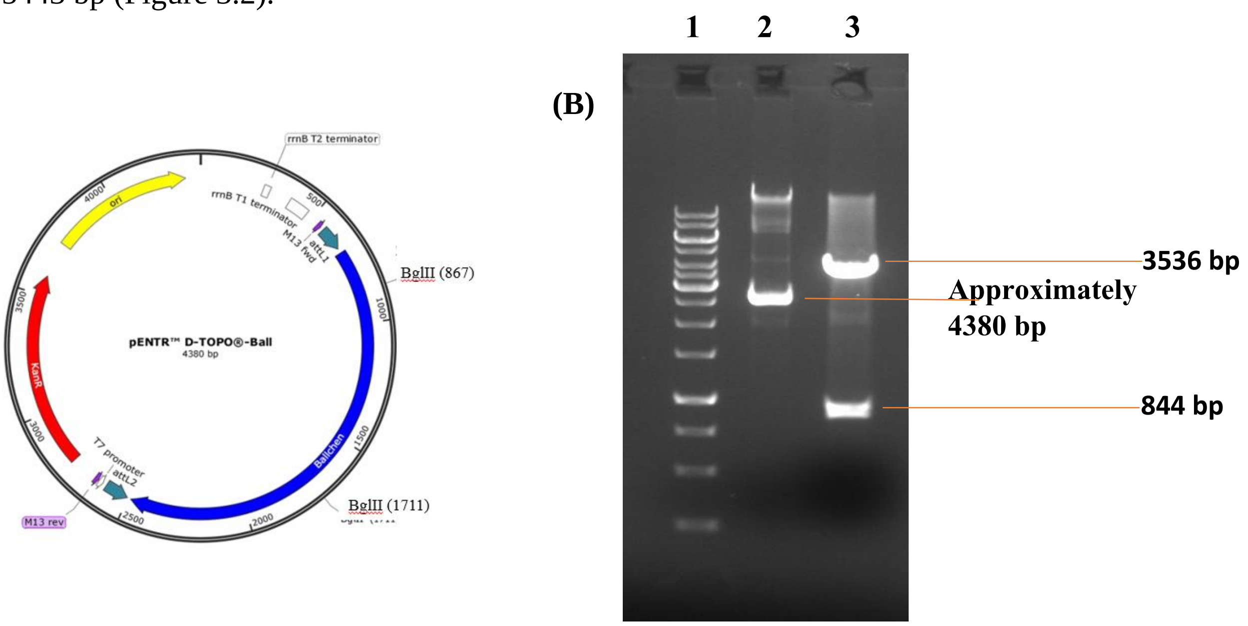


Fig. 3.1: (A) Schematic diagram of the entry vector pENTR™D-TOPO®-ball (B) Restriction digestion with BglII produced two fragments of 3536 bp and 844 bp, confirming successful cloning. The digested plasmid (lane 3) was analyzed alongside the undigested plasmid (lane 2) and a 1 kb DNA marker (lane 1).

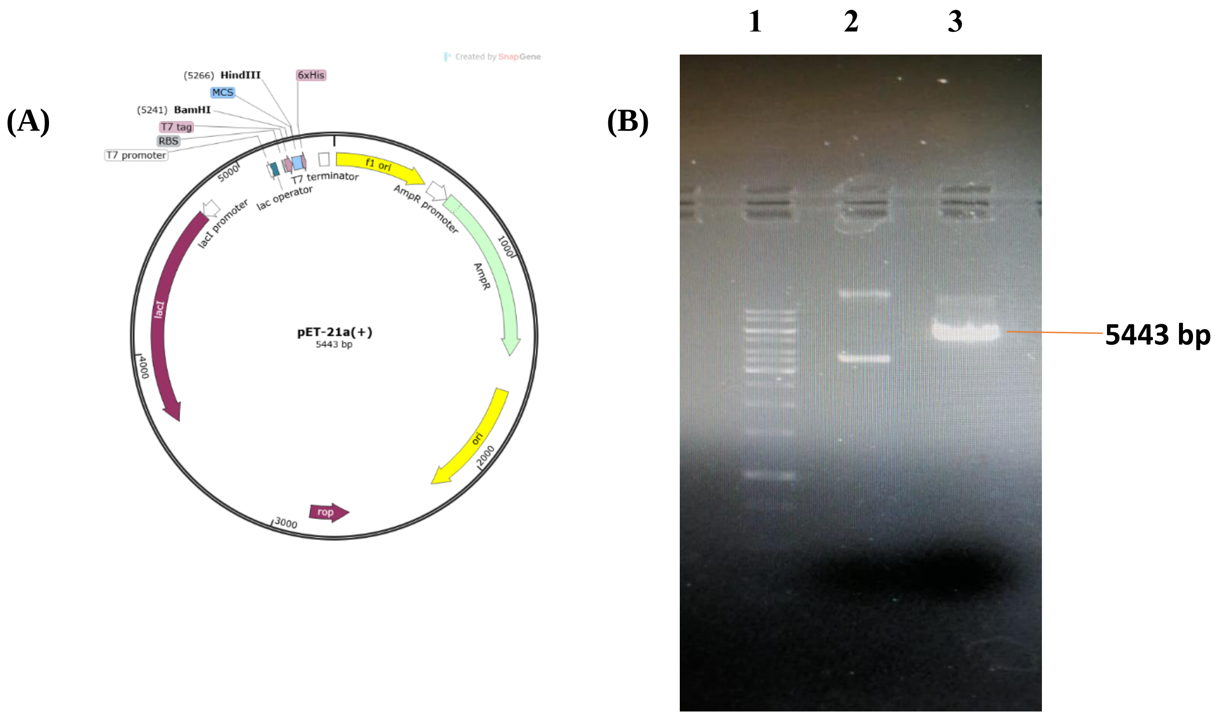


Fig 3.2: (A) Schematic of the pET21a (+) expression vector used for cloning. (B) pET21a(+) Confirmation via restriction digestion; Lane 1: 1 kB DNA ladder; Lane 2: Uncut plasmid shows plasmid in different conformations; Lane 3: pET21a (+) digested with BamH1-HF shows linearized plasmid of 5443 bp

3.2 PCR amplification and cloning of BALL kinase domain

For the amplification of ball kinase coding sequence, initially *Taq Polymerase* was used to have an idea of the best annealing temperature for the given primer sets. After the PCR optimization for amplification of BALL kinase coding sequence, the insert was amplified with the use of Phusion Polymerase which possess proof reading activity.

Due to its efficiency and thermostability at higher temperatures, Taq DNA polymerase is commonly used in molecular biology experiments to carry the PCR. However, in comparison to other polymerases, with proofreading capability, it has a slightly higher rate of error. This is because it lacks a 3′→5′ exonuclease activity, which means it cannot correct errors made during DNA synthesis. The reported error rates for Taq Polymerase range from approximately 1 × 10^−5 to 2 × 10^−4 mutations per base pair per template duplication. Though Taq Polymerase is highly processive and has an extension time of 1 minute per kb (Saiki et al., 1988;

Keohavong et al., 1989; Eckert et al., 1990) but its error rates indicate that Taq polymerase can introduce mutations at a relatively low frequency but still higher than polymerases with proofreading abilities.

For downstream processing of PCR products where they are required for application such as molecular cloning and eventually carrying DNA sequencing, DNA polymerases with proofreading capabilities are preferred to avoid any point mutations. Phusion polymerase was therefore used for cloning purposes. This is an ideal choice for cloning purposes because it ensures fidelity of the amplified PCR product and minimizes any chances of mutation, as it possesses 3→5′ exonuclease proofreading domain in addition to its polymerase activity. This proofreading ability allows them to recognize and correct misincorporated nucleotides during DNA synthesis, resulting in significantly lower error rates compared to Taq polymerase. By minimizing the introduction of errors during amplification, such high-fidelity polymerases contribute to the reliability and accuracy of downstream analyses. Phusion has an extension time of 15 seconds per 1 kb (McInerney et al., 2014).

In addition to this, the Phusion DNA Polymerase comprises of a unique structure which combines a *Pyrococcus*-like enzyme that carries a processivity-enhancing domain (McInerney et al., 2014). This ensures enhanced fidelity and speed. Such structural features allow Phusion to be well-suited for cloning and can handle long or challenging amplicons. Furthermore, *Phusion DNA Polymerase* exhibits an error rate that is more than 50 times lower than that of Taq DNA Polymerase and 6 times lower than *Pyrococcus furiosus* DNA Polymerase (McInerney et al., 2014). Overall, Phusion DNA Polymerase stands out as one of the most accurate thermostable polymerases available (McInerney et al., 2014).

The BALL kinase coding sequence was amplified with Phusion polymerase (Figure 3.3). Since the PCR product was to be used for restriction digestion followed by elution from gel and finally ligation with the expression vector, at least 1 µg amount of DNA was required, for each of the inserts for N and C term fusion constructs. Therefore, PCR was carried in duplicates, for each of the N and C terminal His tagged ball kinase plasmids, and then PCR product from respective primer sets, was pooled together (Fig 3.4). To ensure the removal of enzymes, nucleotides, primers and buffer components, pooled PCR amplified product was purified via PCR purification kit (Figure. 3.5). Subsequently, the pET-21a (+) expression vector and the PCR pooled products were treated with restriction enzymes, BamHI and HindIII, to generate complementary sticky ends to perform ligation as mentioned in section 2.12. The recombinant plasmids were confirmed first using colony PCR (Figure 3.7) and then restriction digestion (Figure 3.8). The confirmation of recombinant plasmids with BamHI and HindIII restriction enzymes resulted in release of expected 846 bp BALL kinase domain coding sequence cloned in pET-21a (+) vector.

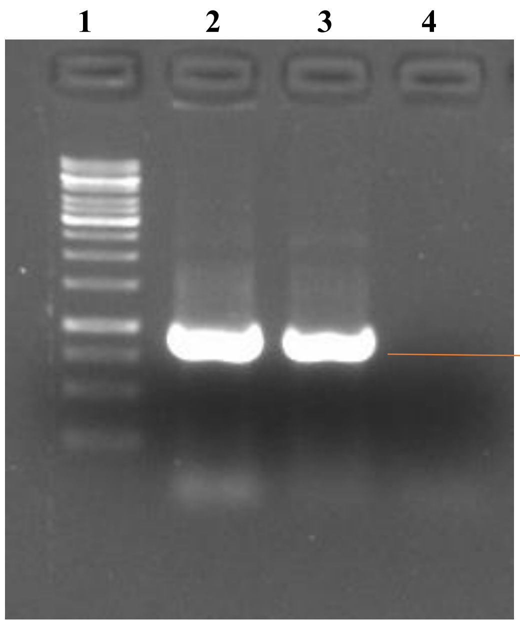


Fig. 3.3: PCR amplified Ball kinase of length 846 base pair Lane 1 shows the 1kB DNA Ladder; Lane2 is N-His Ball Kinase; Lane 3 is C-His Ball Kinase and Lane 4 shows PCR-negative

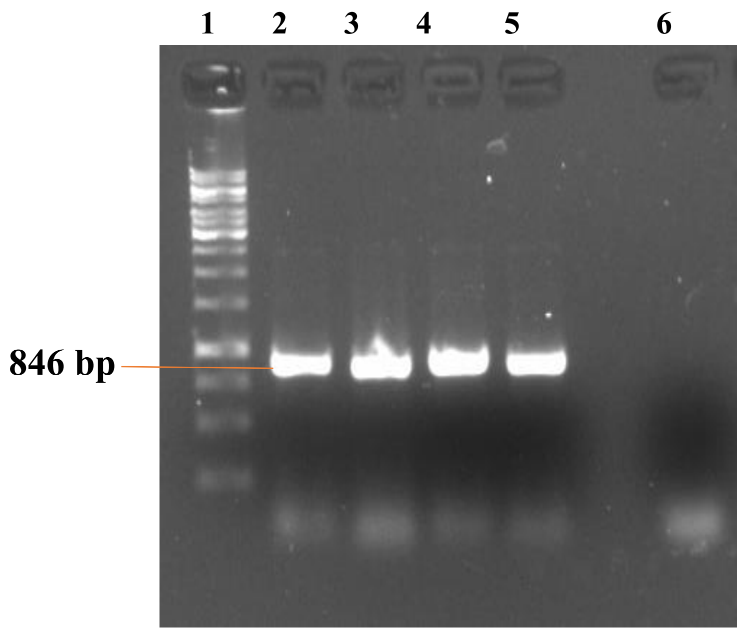


Fig 3.4 PCR Results for amplification N and C His-tagged Ball Kinase in duplicates. Lane 1:1kB DNA Ladder; Lane2: N-His Ball Kinase Sample 1 ; Lane 3:N-His Ball Kinase Sample 2; Lane 4:C-His Ball Kinase Sample 1; Lane 5: C-His Ball Kinase Sample 2 ;Lane 6: PCR negative

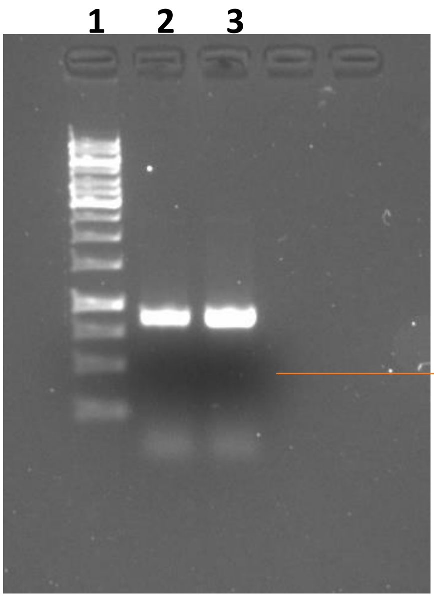


Fig 3.5; PCR purification results for N and C His-tagged Ball Kinase in duplicates; Lane 1: 1kB DNA ladder; Lane 2: N-His Ball Kinase PCR purified product; Lane 3: C-His Ball Kinase PCR purified product

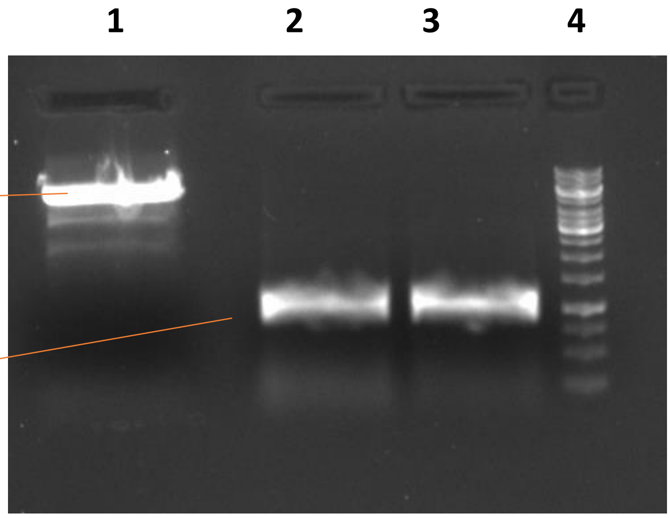
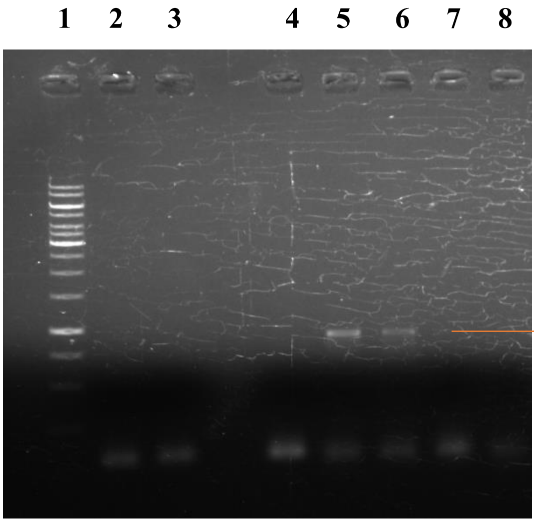


Fig 3.6: Restriction Digestion with BamHI and HindIII of pET-21 a (+), N-His Ball Kinase and C-His Ball Kinase to generate 'sticky ends' Lane 1: pET-21 a (+) double digested with BamHI and HindIII; Lane 2: N-His Ball Kinase double digested with BamHI and HindIII; Lane 3: C-His Ball Kinase double digested with BamHI and HindIII; Lane 4: 1kB DNA ladder

(A)



(B)

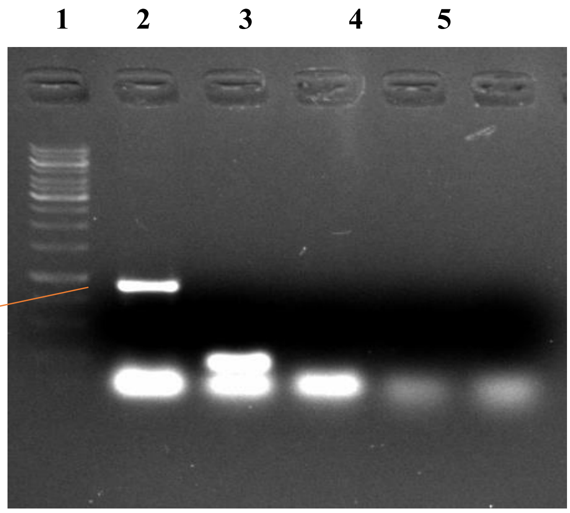


Fig 3.7: (A) Colony PCR results for N and C His tagged BALL kinase domain recombinants. Lane 1:1kB DNA Ladder; Lane 2: N-His Ball Kinase Colony 4 ; Lane 3: N-His Ball Kinase Colony 5 ; Lane 4: N-His Ball Kinase Colony 6 ; Lane 5: C-His Ball Kinase Colony 4 ; Lane 6: C-His Ball Kinase Colony 5; Lane 7: C-His Ball Kinase Colony 6 ; Lane 8: PCR Negative. Expected band of 846 bp is seen for C-His tagged recombinant colonies 4 and 5.

Fig 3.7: (B) Colony PCR results for N His tagged BALL kinase domain recombinants. Lane 1:1kB DNA Ladder; Lane 2: N-His Ball Kinase Colony 7 ; Lane 3: N-His Ball Kinase Colony 8 ; Lane 4: N-His Ball Kinase Colony 9 ; Lane 5: N-His Ball Kinase Colony 10 ; Lane 6: PCR negative. Expected band of 846 bp is seen for N-His tagged recombinant colony 7.

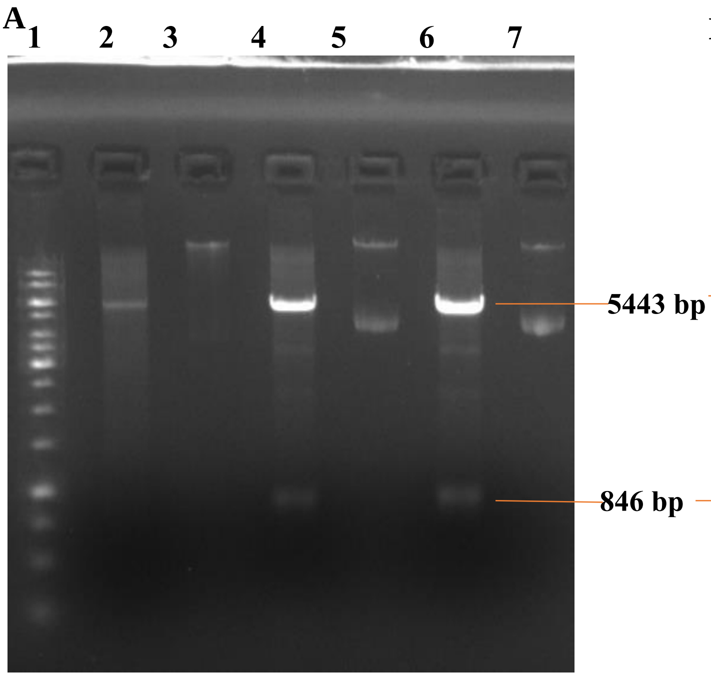


Fig 3.8: (A) Double digestion restriction confirmation for Ball kinase cloned into pET21a(+) vector- The number denotes the original number of colony with reference to transformants; Lane 1:1kB DNA Ladder; Lane 2: Mini N-His Ball Kinase 7 cut ;Lane 3: Mini N-His Ball Kinase 7 uncut ; Lane 4: Midi C-His Ball Kinase 4 cut; Lane 5: Midi C-His Ball Kinase 4 uncut ; Lane 6: Midi C-His Ball Kinase 5 cut; Lane 7: Midi C-His Ball Kinase 5 uncut (The gel shows plasmid amplified via mini prep for N His 7 and midi prep for C His 4 and 5)

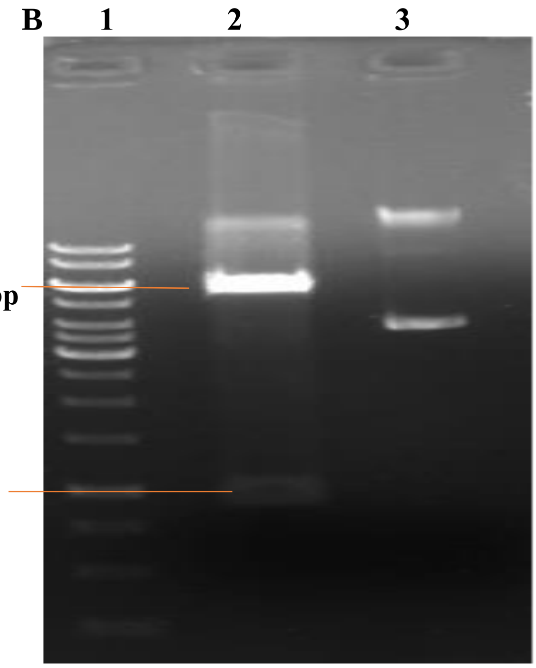


Fig 3.8: (B) Double digestion restriction confirmation for Ball kinase cloned into pET21a(+) vector- The number denotes the original number of colony with reference to transformants; Lane 1:1kB DNA Ladder; Lane 2: Midi N-His Ball Kinase 7 cut; Lane 3: Midi N-His Ball Kinase 7 uncut (The gel shows plasmid amplified via midi prep)

In order to express BALL kinase domain, using, the pET-21a (+) BALL plasmids (N and C term tagged BALL) was initially transformed in the *E.coli* expression strain BL21(DE3). The transformed cells were grown in LB broth and IPTG (1mM) was added to induced expression of BALL when bacterial OD_{600} was 0.5. After 4 Hours of IPTG induction bacterial cells were sonicated to lyse th eclls. Finally, total proteins lysate was resolved on run on SDS gel to visualize expression 31 kDa ball kinase domain protein. In the initial course of experiments, the desired truncated 31 kDa BALL was not observed when compared with uninduced samples. *E.coli* BL21(DE3) has found a common ground in numerous biotechnological applications, particularly in production of recombinant protein. This strain possesses several key characteristics that make it well-suited for its role as an industrial host. Features including rapid cell growth in minimal media, minimal acetate production even when cultivated on high concentrations of glucose, low levels of protease expression, and suitability for high-density culture conditions, make this strain a convenient option for molecular biology experiments. These attributes collectively contribute to BL21(DE3)'s effectiveness as an established workhorse for the efficient production of recombinant proteins in various biotechnological processes (Yoon et al., 2009).

Initially the desired 31kDa BALL kinase domain was not observed in either N or C term tagged BALL, which could be due to presence of induced protein in inclusion bodies or this protein could be toxic to the bacterial cells. To this end, engineered strain BL21(DE3) containing plasmids with the T7 lysozyme gene can be exploited to circumvent the toxic effects of recombinant protein. T7 lysozyme, a natural inhibitor of T7 RNA polymerase, can be regulated using different promoter systems to modulate T7 RNA polymerase activity, thus optimally reducing recombinant protein production rates. This approach helps mitigate the toxic effects associated with protein overexpression, leading to enhanced yields of recombinant proteins (William Studier, 1991). The system ensures the inhibition of protein production of undesired proteins, without IPTG, resulting in higher yields of the desired protein of interest, which is only activated upon IPTG induction. Additionally, the weak promoter ensures that expression levels are tightly controlled, reducing background expression and enhancing the purity of the protein product (Joshi et al., 2005). Furthermore, utilizing this modified strain eliminates the need for lysozyme mediated digestion of induced bacteria for the release of inclusion bodies, which are aggregates of overexpressed proteins commonly formed in conventional protein expression systems. This streamlined process results in the expression of protein with higher purity in comparison to the conventional BL21(DE3) strain, making it an attractive choice for protein expression experiments where high yield and purity are desired (Joshi et al., 2005).

Consequently, the Rosetta (DE3) strain of *E. coli* was used in further experiments. By switching strain from *E. coli* BL21(DE3) to *E. coli* Rosetta (DE3), It was anticipated that the overall success rate of protein production would improve significantly. Rosetta (DE3) *E. coli* strain overcomes the challenges of codon bias through the co-expression of genes encoding rare tRNAs. This engineered strain compensates for the scarcity of certain cordons such as AGG, AGA, AUA, CUA, CCC, and GGA, which are commonly utilized by eukaryotes but are infrequently used in *E. coli*. This would therefore work better for the expression of BALL kinase domain sequence which is indeed a eukaryotic coding sequence. So, by ensuring the supply of the necessary tRNAs for these rare codons, the Rosetta (DE3) strain facilitates the translation of heterologous genes with improved efficiency, overcoming codon bias and enhancing protein expression levels (Gustafsson et al., 2004).

Although the expected 31 kDa BALL domain was observed on SDS gel after induction in Rosetta (DE3) cells these results however were not reproducible. Moreover, expression of BALL kinase domain was only found in the pellet fractions when compared to the soluble fractions from lysate. Although both N term as well as C term tagged proteins were visible in the insoluble fractions but amounts were too low and not reproducible across experiments. After a series of changes in protein expression protocol which include the time for protein induction (from 4 hours to overnight) , the $O.D_{600}$ for IPTG induction (0.5 to 0.8), the concentration of IPTG (0.1 mM to 2 mM), and the temperature (15°C- 37°C) of induction no improvement in protein production was observed.

In addition, chemical lysis method was used to ensure cells are completely lysed. The lysis buffer in 1X PBS comprised of different enzymes to aid in protein extraction from the bacterial cells. The lysozyme was used in the buffer to aid in bacterial cell wall lysis and the benzonase to completely digest nucleic acids. In addition, the lysis buffer contained various protease inhibitors including PMSF, leupeptin, pepstatin and aprotonin. The rationale behind using chemical lysis method for protein expression and islolation was to ensure that the BALL kinase domain was not degraded while collecting protein sample. Using a variety of protease inhibitors would ensure that the BALL kinase is not attacked from bacterial proteases. However, the SDS gel results did not show the desired band of 31kDa corresponding to BALL kinase domain.

3.3 Expression of BALL kinasa domain in BL21 (DE3)-Codon Plus RIL cells

Next, BL21 (DE3)-Codon Plus RIL were used for expression of BALL kinase domain in bacteria. The IPTG concentration used for induction of BALL expression was 1 mM and The cell collection occurred following a 4-hour induction period at 37°C. The total cell lysate was prepared by sonication of bacterial cells in Tris-HCl buffer and protein concentrations were measured using standard Bradford assay. A total of 2.4 ug protein from each sample (uninduced and induced) was resolved on SDS gel. In contrast to uninduced samples, a protein representing BALL kinase domain was clearly seen at expected 31kDa size (Figure 3.10). An important alteration to the protocol was to run the SDS gel for a total of 6 hours. This ensured that the gel was run to completion, allowing the lower protein markers to resolve clearly, making it easier to visualize the marker corresponding to BALL kinase domain. The same protocol was used for large-scale and small-scale protein expression.. However, expression of an ample amount of BALL kinase domain was only visible in pellet fraction of lysate as compared to soluble fraction. It indicates presence of induced protein in inclusion bodies or insoluble fraction. This protein visible in the insoluble fraction i.e., pellet was not present in both the pellet and soluble fraction of uninduced samples.

The expression system using *E.coli* offers several advantages but also presents various challenges, including low expression levels for proteins with higher molecular weight and the production of inclusion bodies. One of the foremost challenges in expression of eukaryotic proteins in bacteria is proper folding, which is particularly for heterologous proteins that contain disulfide bonds. In addition, codon bias is also another frequently encountered obstacle when working with expression of eukaryotic proteins in prokaryotic system, because certain codons are preferred over others. The codon bias can lead to complications in expression efficiency and protein solubility. Rare cordons in *E. coli* include AGG/AGA/CGG (arginine), AUA (isoleucine), CUA (leucine), CCC (proline), and GGA (glycine) (Nouri et al., 2016). Strategies for addressing codon bias include codon optimization through site directed mutagenesis or enhancing the availability of rare tRNAs via genome modification of the host. While codon optimization can be costly, expressing recombinant proteins rich in rare codons in strains that co-express the corresponding tRNAs offers a more economical solution (Gustafsson et al., 2004). In modified *E. coli* strains, such as BL21 Codon Plus (DE3)-RIL, efficient expression of proteins containing high levels of AGG/AGA (arginine), AUA (isoleucine), and CUA (leucine) codons is achieved. Likewise, the Rosetta (DE3) strain, which is genetically altered, possesses tRNAs for a wider array of rare codons (Nouri et al., 2016).

These findings suggest that *E. coli* BL21 Codon Plus (DE3-RIL) is a good choice for expressing recombinant BALL kinase protein, which contains rare codons. BALL kinase contains 105 rare codons which is approximately 37.2% of the total codon count. Out of these 105 rare codons, 2 encode for cysteine residues, 7 for valine and a total of 8 are arginine encoding codons. This strain's ability to efficiently express proteins with specific rare codons makes it particularly well-suited for such applications. While most protein expression protocols, for downstream utilization, such as purification, antibody generation or functional characterization show ideal results with Rosetta strain, this was however not true in the case of BALL kinase, which was expressed in *E. coli* BL21 Codon Plus (DE3-RIL).

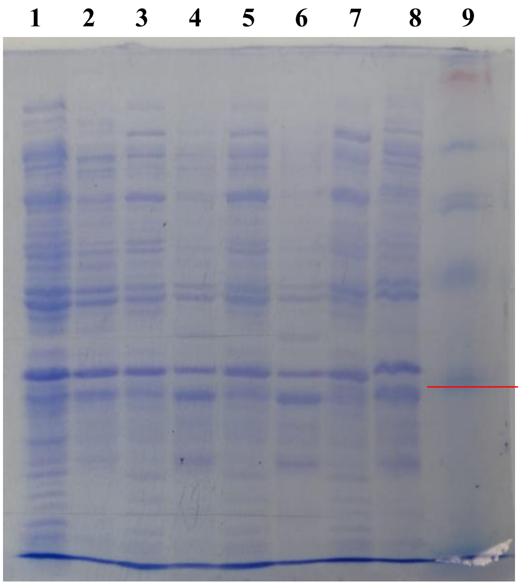


Fig 3.9 Test expression for ball kinase transformed in BL21(DE3) cells with 1mM IPTG and incubation for 4 hours at 37°C. The expected size of band is visible in all samples including uninduced Lane 1: Uninduced supernatant; Lane 2: Uninduced pellet; Lane 3: C-His 5 Supernatant; Lane 4: C-His 5 Pellet; Lane 5: C-His 4 Supernatant; Lane 6: C-His 4 Pellet; Lane 7: N-His 7 Supernatant; Lane 8: N-His 7 Pellet; Lane 9: SeeBlue™ Plus2 Ladder

Ball kinase domain ~31.302 kDa

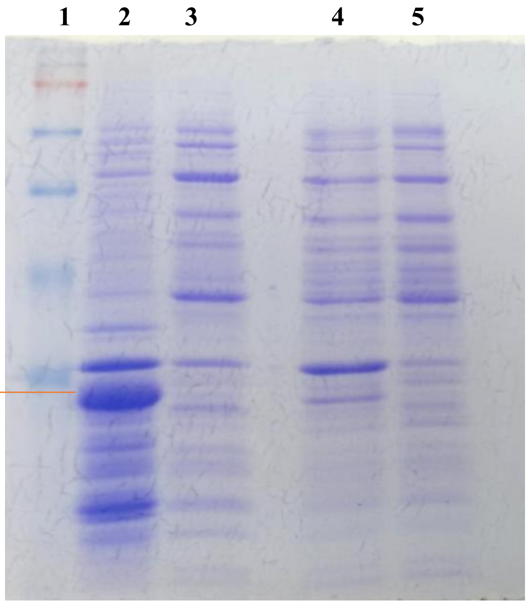


Fig. 3.10 Ball kinase test expression in BL21 (DE3)-Codon Plus RIL. Ball kinase test expression confirmation in small scall culture using BL21 (DE3)-Codon Plus RIL. Cells were induced 1mM IPTG and incubation for 4 hours at 37°C. Clear band of expected size 31.302 kDa is seen in induced pellet Lane 1: SeeBlue™ Plus2 Ladder; Lane 2- C-His tagged Ball kinase-Pellet; Lane 3- C-His tagged Ball kinase-Supernatant; Lane 4- Uninduced Pellet; Lane 5- Uninduced Supernatant

3.4 BALL Kinase domain purification

After successfully optimizing expression of BALL kinase domain, initially the C-term tagged BALL construct was used for expression and purification. Since BALL kinase domain was present only in the insoluble fractions, the protein purification was performed under denaturing conditions. The purification of C his tagged ball kinase domain was carried out as described in section 2.21.

Expressing eukaryotic proteins in *E. coli* frequently results in the formation of insoluble proteins known as inclusion bodies. This is also seen in the case of BALL kinase domain. One reason for this observed fact is that the protein synthesis rate is 10-fold higher in prokaryotes, compared to eukaryotes (Georgiou and Valax, 1999). Eukaryotic proteins are synthesized rapidly in bacterial hosts due to their efficient translation machinery. However, their slower folding kinetics might not match this fast synthesis pace, leading to aggregation issues. In contrast, prokaryotic proteins, which fold faster, are better synchronized with the rapid elongation of the polypeptide chain. The disparity in folding rates may contribute to the challenges of expressing eukaryotic proteins in bacteria. The fast synthesis and slower folding of eukaryotic proteins create a situation where misfolding and aggregation become more likely. Proteins in their non-native state, even smaller ones that fold quickly, tend to aggregate when present at high concentrations. This is because their hydrophobic surfaces, normally facing towards inner core in the native state, get exposed in the unfolded state. These exposed hydrophobic regions interact with each other, leading to aggregation. To mitigate this issue, various strategies such as co-expression with molecular chaperones, optimization of growth conditions, or employing eukaryotic hosts for expression can be explored. These strategies aim to balance the synthesis and folding rates, facilitating proper folding and preventing aggregation of eukaryotic proteins in bacterial hosts (Widmann and Christen, 2000). Therefore, for BALL kinase domain a variety of expression hosts with altered conditions including incubation temperature were used. This is also discussed in section 3.3.

The major drawback of such a scenario where protein accumulates in insoluble fraction is that it must be extracted via denaturing agents, and then allowed to fold back into native conformation. For small single-domain proteins (10 to 20 kDa), aggregation is typically not a significant issue, although overall only 5 to 20 percent of those observed for similar proteins expressed in a soluble state may be recovered. However, for larger multiple-domain proteins (ranging between 40 and 70 kDa), recovery is extremely challenging, due to aggregation. Nonetheless, there have been successful scenarios, such as in the case of 69 kDa tissue plasminogen activator protein, where expression in a soluble state has been achieved (Grunfeld et al., 1992).

BALL kinase domain, which was found to be accumulate in insoluble fraction of the total protein, therefore, it was purified under denaturing conditions with urea. Through electrostatic interaction, urea mainly reaches the first solvation shell of the protein and interacts with the protein backbone. Following this, the equilibrium is weighed toward denatured state, by destabilizing the contact between protein molecules (Das and Mukhopadhyay, 2009). In pure water, interactions between protein molecules are more stable than those between protein and water molecules. However, as denaturation advances, water penetrates the protein along with urea, which then effectively solvates various subunits of the protein. Urea enters the protein core by solvating the protein backbone. It forms hydrogen bonds with hydrophilic amino acids within protein and creates dispersion by repelling the hydrophobic amino acids. These processes result in the protein's denaturation (Das and Mukhopadhyay, 2009).

The BALL kinase domain was purified with the Ni-NTA protein purification column that is based on the principle of metal affinity chromatography. Metal-affinity chromatography has been used for the purification of recombinant proteins (Porath et al., 1975). The principle of the technique exploits the exposed histidine or cysteine residues in a protein, can bind to transition metals such as $Ni^{2+,}$ Cu^{2+}, Zn^{2+} (Hochuli et al., 1987). The optimal design of column is where the tetradentate nitrilotriacetic acid (NTA) binds to Ni^{2+} with three carboxyl groups and one nitrogen atom. Whereas on the protein side, six histidine residues in tandem, were enough to specifically bind to Ni^{2+}. Nickel columns are commonly utilized for immobilized metal affinity chromatography (IMAC) in purifying recombinant proteins tagged with polyhistidine, often a hexahistidine tag (6xHis tag or His6 tag). The most common tag is a hexa-histidine. The dissociation constant (KD) of 6 His-tagged protein to Ni–NTA is 10^-13 at pH 8 (Schmitt et al., 1993).

Since there develops an ionic interaction between Ni^{2+} and nitrogen in the imidazole ring of histidine, this binding affinity can be altered by protonating these nitrogen atoms. The pKa of histidines in ribonuclease was determined to be 6.5 (Means and Feeney, 1971). Standard elution conditions for removing a His-tagged protein from a nickel column involve a pH range of 4.5–4.8 or below 5 (Schmitt et al., 1988, 1993), or a gradient from pH 6.0 to 4.0 (Paborsky et al., 1996).

Therefore, to bind, wash and elute the BALL kinase domain under denaturing conditions, a higher concentration of urea in the range 7-8 M was used. In addition, the purification column was sequentially loaded with binding, washing and eluting buffers, gradually reducing pH from 8.0 to 4.5 in the final elution, to allow release of His tagged BALL kinase from the column. The BALL kinase domain was confirmed to be present in the elution fractions (Figure 3.11). One of the challenges while purifying His tagged proteins is that of nonspecific binding with the column, which was also observed in case of BALL kinase domain (Figure 3.11). One possible explanation for this observation is the binding of histidine residues due to bacterial native proteins.

After the BALL kinase domain was eluted, the elution was moved to dialysis tubing, which had a molecular cut off of approximately 12 kDa. This ensured that the BALL kinase domain, of weight 31 kDa would be retained back in the tubing. Additionally, this would allow to dialyze out the urea, down its concentration gradient into the refolding buffer. The process was carried over 3 days time to ensure very slow removal of urea from the protein sample, to allow it to refold back into its native state. The refolding buffer contained ingredients conducive to protein folding. These included salts, buffering agents, reductants and chelating agents such as EDTA. All these were necessary to provide an environment that would support protein folding into native state by maintaining appropriate ionic strength, pH and ensuring any protease remains chelated (Rinas and Vallejo, 2004). It was however noted that the amount protein was lost after the dialysis or most likely the protein was being precipitated (Figure 3.12). Therefore, the dialysis step needs to be optimized. One possible step would be to carry dialysis for even longer time period.

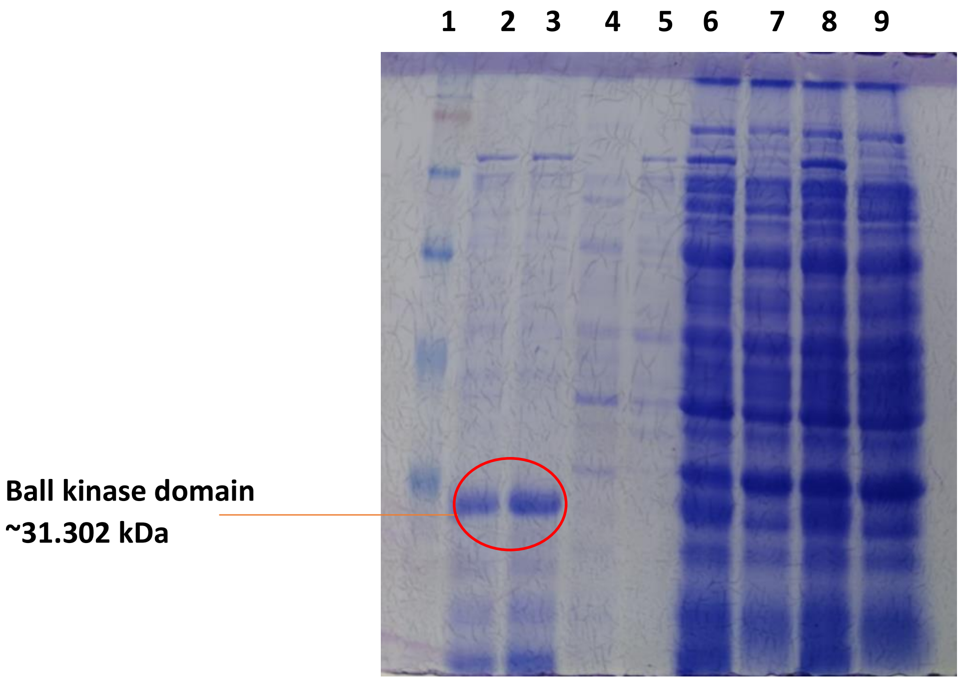


Fig. 3.11 His tagged ball kinase purification SDS gel result. The gel shows a clear band of expected size approximately 31 kDa corresponding ball kinase in Elution 1 and Elution 2. Lane 1: SeeBlue™ Plus2 Ladder; Lane 2: Elution 1, Lane 3: Elution 2; Lane 4; Wash 1; Lane 5: Wash 2; Lane 6: Supernatant induced; Lane 7: Supernatant uninduced; Lane 8: Whole Cell Lysate induced; Lane 9: Whole Cell Lysate uninduced

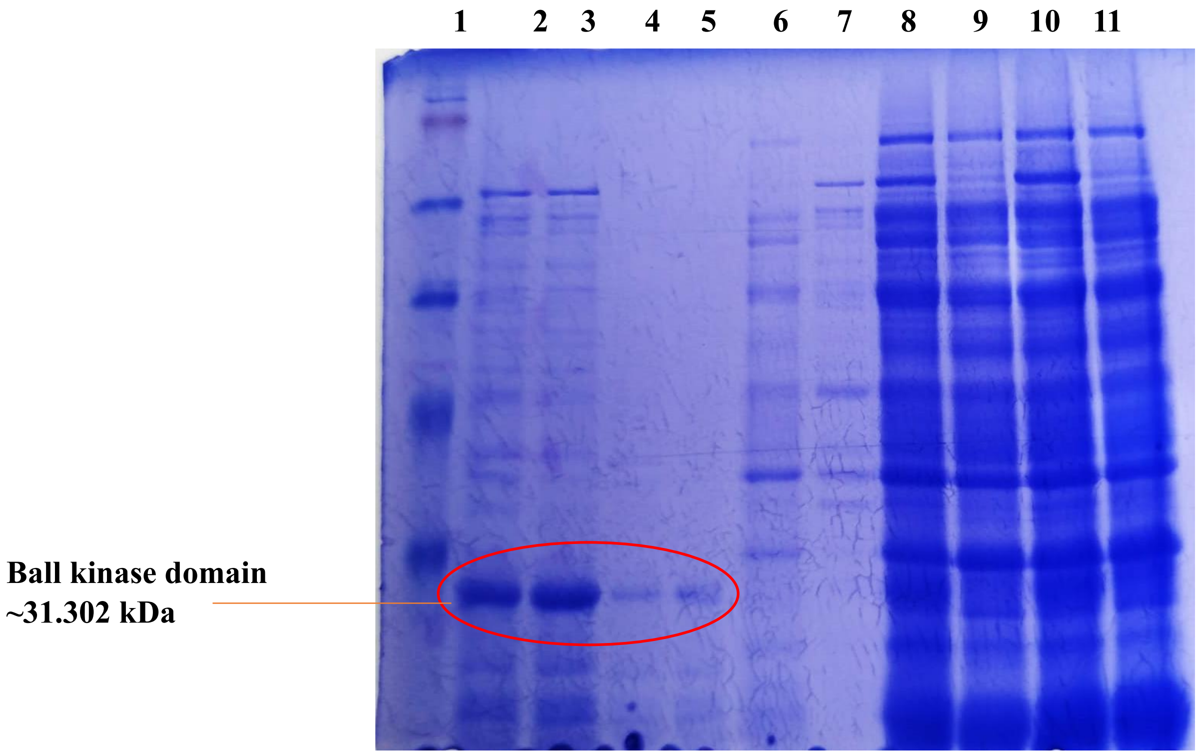


Fig. 3.12 His tagged ball kinase purification through dialysis- SDS gel result. The gel shows a clear band of expected size approximately 31 kDa corresponding ball kinase in Elution 1 and Elution 2 and Dialysed Elution. Lane 1: SeeBlue™ Plus2 Ladder; Lane 2: Elution 1, Lane 3: Elution 2; Lane 4: Dialyzed E1; Lane 5: Dialyzed E2; Lane 5; Wash 1; Lane 6: Wash 2; Lane 7: Supernatant induced; Lane 8: Supernatant uninduced; Lane 10: Whole Cell Lysate induced; Lane 11: Whole Cell Lysate uninduced

Chapter 4: Outlook and Future Perspectives

This study was focused on the basic molecular cloning, expression and purification of the Ballchen kinase domain of *Drosophila*. The overarching theme of the study was to establish an *in vitro* kinase assay that could be used to screen the kinase for putative inhibitors. The repeated steps throughout experiments with slight modifications led to the protocol's optimization. This study presents a robust protocol for the classic cloning, expression and purification of kinase domain. This becomes even more important, given that the kinase domain is highly conserved across species. In fact, VRK1, the ortholog of BALL in humans, demonstrates a highly conserved kinase domain with the BALL. This highly conserved domain across different species indicates it is functionally important and hence kept intact as a part of the structure of the whole protein. Therefore, this protocol should ideally be a good starting point for working with other kinases, in house. This study holds significance as it presents an optimized protocol for kinase domain purification, which can readily be used to set up kinase assay. Kinase assay acts as the stepping stone for screening and identifying the small molecule inhibitors. With VRK1's kinase domain highly conserved with Ball's kinase domain, we can narrow down the hits and pin down the putative inhibitors. In addition, ball kinase inhibitors can then be tested *in vivo* using the fruit fly, *Drosphila Melanogaster*, and RNA level or protein level alterations could be detected upon perturbation by the inhibitors. Furthermore, since abnormal functioning of VRK1 is known to be involved in various types of cancers, the inhibitors could help in drug discovery, targeted for VRK1. As an immediate extension to the project, molecular docking studies using *in silico* tools available online can be used to gain insight into the predicted small molecule inhibitors for BALL kinase.

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Appendices

Appendix 1: Restriction digestion of pENTR-Ball

Reagents	**Amounts (1X)**
10X Buffer O	2 µL
DNA (pENTR-Ball)	1 µg
Bgl11	1µL
Nuclease free water	Up to 20 µL

Appendix 2: Solutions for bacterial culture

The following amounts were poured into sterilized 1000 mL glass bottle and prepared media was sterilized by autoclaving at at 15 psi and 121°C for 25 minutes.

LB Broth

Reagents	Volume
Trypton	5 g
NaCl	2.5 g
Yeast	2.5 g
Distilled Water	Upto 500 mL

LB Agar

Reagents	Volume
Trypton	5 g
NaCl	2.5 g
Yeast	2.5 g
Agar	7.5 g
Distilled Water	Upto 500 mL

Appendix 3: Plasmid isolation solutions

P1- Resuspension Buffer

Reagents	Concentration
Tris-Cl	50 mM
EDTA	10 mM
RNase A	100 µg / ul

The buffer's pH was adjusted to 8 using concentrated HCl, and then it was sterilized using an autoclave. After cooling, RNAse A was introduced, and the buffer was kept at 4ºC for storage.

P2- Lysis Buffer

Reagents	Concentration
NaOH	200 mM
SDS	1% w/v

The buffer should not undergo autoclaving and should be stored at room temperature.

P3- Lysis Buffer

Reagents	Concentration
Potassium Acetate	3 M

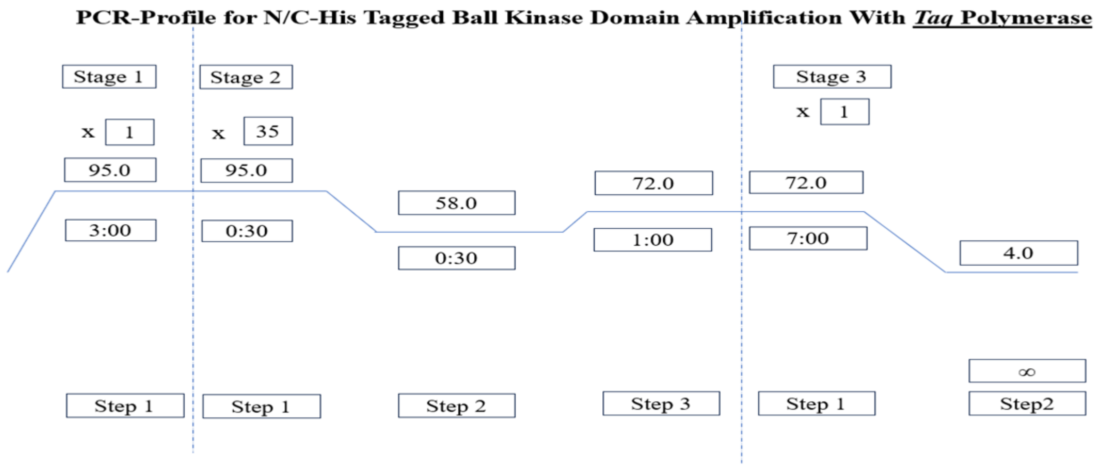
The pH of this buffer was adjusted to 5.5 using glacial acetic acid, and then it was autoclaved. Afterward, the buffer was stored at 4ºC.

Appendix 4: Primer Sequences

	Name	Sequence	Tm
N-His	Ball-CDS-N-His-F	TAGGATCCCATCATCACCATCACCACATGCCGCGTGTAGCC	71.5 °C
	Ball-CDS-R-ws	CTTTAAGCTTCTATCCCTGGTATTT	52.8 °C
C-His	Ball-CDS-F	TACGGATCCATGCCGCGTGTAGCC	62.5 °C
	Ball-CDS-R-ns	CTTAAGCTTTCCCTGGTATTTCCG	55.7 °C
N-His	Ball-Kinase-N-His-F	TACGGATCCATGCATCATCACCATCACCACTGGCGCATTGGACCC	72.7 °C
	Ball-Kinase-R-ws	GCAAAGCTTCTAGAACCAACTGCGACA	59.7 °C
C-His	Ball-Kinase-F	TACGGATCCATGTGGCGCATTGGACCC	64.3 °C
	Ball-Kinase-R-ns	CTTAAGCTTGAACCAACTGCGACA	55.7 °C

Appendix 5: PCR reaction and profile for Ball Kinase using *Taq Polymerase*

Reagents	**Amounts (1X)**
Taq Buffer (NH4SO4)	1.00 µL
DNTPs	0.25 µL
MgCl2	1.25 µL
Forward Primer	0.25 µL
Reverse Primer	0.25 µL
Taq Polymerase	0.15 µL
Template	1.00 µL
PCR water	Upto 10 µL



Appendix 6: Composition of Agarose gel solution used

50X TAE Buffer (make volume up to 100 ml by dissolving in water):

- Tris Base: 24.2 g
- Glacial Acetic Acid: 5.71 ml
- 0.5M EDTA (pH 8.0): 10 ml

Preparation of 0.5M EDTA (pH 8.0):

- Prepare a total volume of 50 ml.
- Using the unitary method, add 9.305 g of disodium EDTA.2H2O to 40 ml of water.
- Add NaOH pellets, stir the solution on a hot plate, and adjust the pH to 8.0.

1% Agarose Gel:

- Weigh 1 g of agarose and dissolve it in 100 ml of 1X TAE buffer by heating in a microwave oven.

6X DNA Loading Dye:

- Bromophenol blue: 25 mg
- Glycerol: 3 ml
- Make up the solution to 10 ml using water.

Appendix 7: PCR reaction and profile for Ball Kinase using Phusion DNA Polymerase

Reagents	Amounts (1X)
HF Buffer	4.00 µL
10mM DNTPs	0.40 µL
Forward Primer	0.50 µL
Reverse Primer	0.50 µL
Phusion DNA Polymerase	0.20 µL
Template	2.00 µL
PCR water	Upto 20 µL

Appendix 8: Restriction Digestion of pET21a (+)

Reagents	Amounts (1X)
DNA	1 µg
10x rCut Smart Buffer	5 µL
BamH1-HF	1.0 µL
Nuclease Free Water	Make upto 25 µL

Appendix 9: Restriction digestion of pET21a (+), N and C His tagged Ball kinase

pET21a (+)

Reagents	Amounts (1X)
10x Tango Buffer	4 µL
DNA- pET21a (+)	1 µL (1µg)
BamH1	1 µL
Hind111	1 µL
PCR Water	Up to 20 µL

N-His tagged Ball kinase

Reagents	Amounts (1X)

10x Tango Buffer	4µL
DNA- N-His Ball kinase	14.36 µL (1µg)
BamH1	1 µL
Hind111	1 µL
PCR Water	Up to 20 µL

C-His tagged Ball kinase

Reagents	Amounts (1X)
10x Tango Buffer	4µL
DNA- C-His Ball kinase	13.8 µL (1µg)
BamH1	1 µL
Hind111	1 µL
PCR Water	Up to 20 µL

Appendix 10: Ligation reaction for N and C His tagged Ball kinase with pET21a (+)

Reagents	Amounts (1X)	Amounts (1X)
T4 DNA ligase	1 µL	1 µL
DNA ligase (10x)	2 µL	2 µL
PET21a (+)	2 µL	2 µL
N-His	1 µL	-
C-His	-	1 µL
PCR water	Upto 20 µL	Upto 20 µL

Appendix 11: Colony PCR for N and C His tagged ball kinase

Reagents	Amounts (1X)
Taq Buffer (NH4SO4)	1.00 µL
MgCl2	1.25 µL
Forward Primer	0.25 µL
Reverse Primer	0.25 µL
Taq Polymerase	0.15 µL
Template (From colony PCR)	1.00 µL
PCR water	Upto 10 µL

Appendix 12: Confirmation of transformed inserts through restriction digestion (the number denotes the colony numbered on the LB agar plate after transformation)

Reagents	Amounts (1X)
10X Tango Buffer	4 µL
DNA (C-His 4)	1.64 µL
Hind111	1.00 µL
BamhH1	1.00 µL
PCR Water	Upto 20 µL

Reagents	Amounts (1X)
10X Tango Buffer	4.00 µL
DNA (C-His 5)	1.45 µL
Hind111	1.00 µL
BamhH1	1.00 µL
PCR Water	Upto 20 µL

Reagents	Amounts (1X)
10X Tango Buffer	4.00 µL
DNA (N-His 7)	2.19 µL
Hind111	1.00 µL
BamhH1	1.00 µL
PCR Water	Upto 20 µL

Appendix 13: SDS- 15 % Resolving Gel Composition

Reagents	Amounts
Distilled Water	2.8 ml
Acrylamide (40%)	3.0 ml
1.5M Tris-HCl (pH 8.0)	2.0 ml
10% SDS	80 µL
10% APS	80 µL
TEMED	8 µL

Stacking Gel (5 mL)

Reagents	Volume
Distilled Water	3.645 ml
Acrylamide (40%)	0.625 ml
1.5M Tris-HCl (pH 8.0)	0.63 ml
10% SDS	50 µL
10% APS	50 µL
TEMED	5 µL

Appendix 14: Composition of SDS solutions used

SDS Running Buffer:

- Glycine: 14.4 g
- Tris: 3.03 g
- SDS: 1.00 g
- Up to 1 L water

SDS Staining Solution (Make final volume up to 1L in water):

- Methanol: 440 ml
- Acetic Acid: 60 ml
- Coomassie Brilliant Blue R-250: 2.25 g

SDS Destaining Solution (1L):

- Methanol: 300 ml
- Acetic Acid: 100 ml
- Distilled water: 600 ml

10% APS Preparation:

- Add APS powder to an Eppendorf tube and note its weight in milligrams.
- Add water in an amount equal to the weight in milligrams multiplied by 10, in microliters, and dissolve.

Acrylamide was provided and used as such.

4X Laemmli Buffer:

- 1M Tris-Cl, pH 6.8: 10 ml
- 50% Glycerol: 20 ml
- 10% SDS: 5.0 g
- DTT: 10 ml
- 1% Bromophenol blue: 0.1 g
- Distilled water: Up to 50 ml

Preparation of 1M Tris HCl:

- Add 12.11 g Tris to a 500 ml glass bottle.
- Make up the volume to 100 ml by adding water and dissolve.
- Adjust the pH to 6.8 using a pH meter.

Preparation of 1.5M Tris HCl:

- Add 18.15 g Tris to a 500 ml glass bottle.
- Make up the volume to 100 ml by adding water and dissolve.
- Adjust the pH to 8.8 using a pH meter.

Preparation of 10% SDS Solution:

- Dissolve 10 g SDS in 80 ml water.
- Make the final volume up to 100 ml.

1M DTT:

Dissolve 1.5 g DTT in 10 ml water to make 1M DTT.

Appendix 15: Solutions for protein purification with Ni-NTA Column

Buffer B (adjust pH to 8.0)

Reagents	Volume
100mM NaH_2PO_4	6.90 g $NaH_2PO_4.H_2O$
10mM Tris-Cl	0.61 g Tris Base
7 M Urea	210.21 g
Distilled Water	Upto 1000 mL

Buffer E (adjust pH to 4.5)

Reagents	Volume
100mM NaH_2PO_4	6.90 g $NaH_2PO_4.H_2O$
10mM Tris-Cl	0.61 g Tris Base
8 M Urea	240.24 g
Distilled Water	Upto 1000 mL

Buffer C (adjust pH to 6.3)

Reagents	Volume
100mM NaH_2PO_4	6.90 g $NaH_2PO_4.H_2O$
10mM Tris-Cl	0.61 g Tris Base
7 M Urea	210.21 g
Distilled Water	Upto 1000 mL

Appendix 16: Folding buffer composition for dialysing ball kinase

1X PBS

Reagents	Volume
NaCl	8 g
KCl	0.2 g1
Na_2HPO_4	1.44 g
KH_2PO_4	0.24 g
Distilled Water	Upto 1000 mL

Folding Buffer A (Dissolve the complete reagents in 1L PBS and store at 4C)

Reagents	Volume
2 M Glycine Stock	172 mL
2 M NaOH Stock	7.8 mL
Sucrose	100 g
0.5M EDTA pH 8.0	2 mL
Glutathione (Reduced)	310 mg

LAHORE UNIVERSITY OF MANAGEMENT SCIENCES

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To Whom It May Concern

The document entitled "**Molecular cloning, expression and purification of Ballchen kinase domain**" submitted by **Mujtaba Ahmed;** Roll # 22140008 of the **Department of Life Sciences, Syed Babar Ali School of Science and Engineering** was checked through the Turnitin on May 20, 2024, to determine if it is plagiarism-free or otherwise. After excluding quoted material and bibliographies, the originality report indicates that the similarity Index is 19 **(Nineteen)** percent that:

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